

This appendix contains (in concise form) parts of probability calculus and statistical inference that are used in this book but may not be sufficiently covered in introductory courses or textbooks. Apart from some univariate distributions, this appendix mainly considers multivariate random variables, as well as likelihood and Bayesian inference for multidimensional parameters.

B.1 Some Univariate Distributions

Definition B.1 Binomial Distribution

A discrete random variable X is said to have a binomial distribution with parameters $n \in \{1, 2, \dots\}$ and $\pi \in [0, 1]$ if it has (discrete) probability function

$$f(x) = \binom{n}{x} \pi^x (1 - \pi)^{n-x}, \quad x = 0, 1, \dots, n.$$

The mean and variance are given by

$$E(X) = n\pi,$$

$$\text{Var}(X) = n\pi(1 - \pi).$$

We write $X \sim B(n, \pi)$. For $n = 1$ we also speak of a Bernoulli distribution.

A binomial distributed random variable results as the sum of independent and identically distributed binary random variables $X_1, \dots, X_n$ with $P(X_i = 1) = \pi$ and $P(X_i = 0) = 1 - \pi$. Then

$$X = X_1 + \dots + X_n \sim B(n, \pi).$$

Note that the independence of the X_i is crucial for this result.

Definition B.2 Beta Distribution

A continuous random variable X is said to have a beta distribution with parameters $a > 0$ and $b > 0$ if it has probability function

$$f(x) = \frac{\Gamma(a+b)}{\Gamma(a)\Gamma(b)} x^{a-1} (1-x)^{b-1}, \quad x \in (0, 1),$$

where $\Gamma(\cdot)$ is the gamma function. The mean and the variance are given by

$$\begin{aligned} E(X) &= \frac{a}{a+b}, \\ \text{Var}(X) &= \frac{ab}{(a+b)^2(a+b+1)}. \end{aligned}$$

We write $X \sim \text{Beta}(a, b)$. For $a = b = 1$, we obtain a uniform distribution in the interval $(0, 1)$.

Definition B.3 Beta-Binomial Distribution

A discrete random variable X is said to have a beta-binomial distribution with parameters $n \in \{1, 2, \dots\}$, $a > 0$, $b > 0$ if it has probability function

$$f(x) = \frac{\Gamma(a+b)}{\Gamma(a)\Gamma(b)\Gamma(a+b+n)} \binom{n}{x} \Gamma(a+x)\Gamma(a+n-x) \quad x = 0, 1, 2, \dots, n.$$

The mean and the variance are given by

$$\begin{aligned} E(X) &= n \frac{a}{a+b}, \\ \text{Var}(X) &= n \frac{ab}{(a+b)^2} \frac{a+b+n}{a+b+1}. \end{aligned}$$

We write $X \sim \text{BetaB}(n, a, b)$. For $a = b = 1$, the beta-binomial distribution corresponds to a discrete uniform distribution on $0, 1, \dots, n$, i.e., $f(x) = 1/(n+1)$.

The beta-binomial distribution arises as a mixture distribution. Suppose $X | \pi \sim B(n, \pi)$ and $\pi \sim \text{Beta}(a, b)$, then $X \sim \text{BetaB}(n, a, b)$.

Definition B.4 Poisson Distribution

A discrete random variable X is said to have a Poisson distribution with parameter $\lambda > 0$ if it has probability function

$$f(x) = \frac{\lambda^x}{x!} \exp(-\lambda), \quad x = 0, 1, 2, \dots$$

The mean and the variance are given by

$$E(X) = \lambda,$$

$$\text{Var}(X) = \lambda.$$

We write $X \sim \text{Po}(\lambda)$.

Definition B.5 Normal Distribution and Truncated Normal Distribution

A continuous random variable X is said to have a normal (or Gaussian) distribution if it has probability density function (p.d.f)

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right).$$

The mean and variance are given by

$$E(X) = \mu,$$

$$\text{Var}(X) = \sigma^2.$$

We write $X \sim N(\mu, \sigma^2)$. If $\mu = 0$, $\sigma^2 = 1$, then X is said to have a standard normal distribution.

If X is restricted to $a \leq X \leq b$, then it is said to have a truncated normal distribution. We write $X \sim \text{TN}_{a,b}(\mu, \sigma^2)$. Its p.d.f. is given by

$$g(x) = \begin{cases} \frac{f(x)}{P(a \leq X \leq b)}, & a \leq x \leq b, \\ 0, & \text{else.} \end{cases}$$

The support of X is restricted to the interval $[a, b]$ and the density has to be renormalized. For $a = -\infty$ or $b = \infty$, X is said to be left or right truncated.

Definition B.6 Lognormal Distribution

A continuous nonnegative random variable X is said to have a lognormal distribution if $Y = \log(X)$ follows a $N(\mu, \sigma^2)$ distribution. Its p.d.f. is

$$f(x) = \frac{1}{\sqrt{2\pi}\sigma} \frac{1}{x} \exp\left(-(\log(x) - \mu)^2 / 2\sigma^2\right), \quad x > 0.$$

The mean and variance are given by

$$E(X) = \exp(\mu + \sigma^2/2),$$

$$\text{Var}(X) = \exp(2\mu + \sigma^2) \cdot (\exp(\sigma^2) - 1).$$

We write $X \sim \text{LN}(\mu, \sigma^2)$.

Definition B.7 Inverse Gaussian Distribution

A continuous nonnegative random variable X is said to have an inverse Gaussian distribution with parameters $\mu > 0$ and $\lambda > 0$ if it has p.d.f.

$$f(x) = \left(\frac{\lambda}{2\pi x^3} \right)^{\frac{1}{2}} \exp \left(-\frac{\lambda(x - \mu)^2}{2\mu^2 x} \right) \quad x > 0.$$

The mean and variance are given by

$$E(X) = \mu,$$

$$\text{Var}(X) = \mu^3 / \lambda.$$

We write $X \sim \text{invGauss}(\mu, \lambda)$.

Definition B.8 Laplace Distribution

A continuous random variable X is said to have a Laplace distribution (or double exponential distribution) with location parameter μ and scale parameter $s > 0$ if it has p.d.f.

$$f(x) = \frac{1}{2s} \exp \left(-\frac{|x - \mu|}{s} \right).$$

The mean and variance are given by

$$E(X) = \mu,$$

$$\text{Var}(X) = 2s^2.$$

We write $X \sim \text{La}(\mu, s)$.

Definition B.9 Gamma Distribution

A continuous nonnegative random variable X is said to have a gamma distribution with parameters $a > 0$ and $b > 0$ if it has p.d.f.

$$f(x) = \frac{b^a}{\Gamma(a)} x^{a-1} \exp(-bx), \quad x > 0.$$

The mean and variance are given by

$$E(X) = a/b,$$

$$\text{Var}(X) = a/b^2.$$

The mode is $(a - 1)/b$ (for $a > 1$). We write $X \sim G(a, b)$.

An alternative parameterization of the p.d.f., depending on $\mu = E(X)$ and a scale parameter $\nu > 0$, is

$$f(x) = \frac{1}{\Gamma(\nu)} \left(\frac{\nu}{\mu} \right)^\nu x^{\nu-1} \exp\left(-\frac{\nu}{\mu}x\right), \quad x > 0.$$

This alternative p.d.f. is used, for example, in Chap. 5 for gamma regression models.

Definition B.10 Exponential Distribution

A continuous nonnegative random variable X is said to have an exponential distribution with parameter $\lambda > 0$ if it has p.d.f.

$$f(x) = \lambda \exp(-\lambda x), \quad x > 0.$$

The mean and variance are given by

$$E(X) = 1/\lambda,$$

$$\text{Var}(X) = 1/\lambda^2.$$

We write $X \sim \text{Expo}(\lambda)$. The exponential distribution is a special gamma distribution with $a = 1$, $b = \lambda$.

Definition B.11 Chi-Squared Distribution

A continuous nonnegative random variable X is said to have a chi-squared distribution with n degrees of freedom if it has p.d.f.

$$f(x) = \frac{1}{2^{\frac{n}{2}} \Gamma(\frac{n}{2})} x^{\frac{n}{2}-1} \exp\left(-\frac{1}{2}x\right), \quad x > 0.$$

The mean and variance are given by

$$E(X) = n,$$

$$\text{Var}(X) = 2n.$$

We write $X \sim \chi_n^2$. The chi-squared distribution is a special gamma distribution with $a = n/2$ and $b = 1/2$.

If $X_1, \dots, X_n$ are i.i.d. $N(0, 1)$ variables, then

$$Y_n = \sum_{i=1}^n X_i^2$$

is χ_n^2 -distributed. By the (strong) law of large numbers, this representation implies $Y_n/n \rightarrow 1$ for $n \rightarrow \infty$ (almost surely).

Definition B.12 Inverse Gamma Distribution

If $Y \sim G(a, b)$, then $X = 1/Y$ has an inverse gamma distribution with p.d.f.

$$f(x) = \frac{b^a}{\Gamma(a)} x^{-(a+1)} \exp(-b/x), \quad x > 0.$$

The mean and variance are given by

$$E(X) = b/(a - 1), \quad a > 1,$$

$$\text{Var}(X) = b^2/((a - 1)^2(a - 2)), \quad a > 2.$$

We write $X \sim \text{IG}(a, b)$.

Definition B.13 t-Distribution

A continuous random variable X is said to have a t -distribution with n degrees of freedom if it has p.d.f.

$$f(x) = \frac{\Gamma(n+1)/2}{\sqrt{n\pi} \Gamma(n/2) (1 + x^2/n)^{(n+1)/2}}.$$

The mean and variance are given by

$$E(X) = 0, \quad n > 1,$$

$$\text{Var}(X) = n/(n - 2), \quad n > 2.$$

We write $X \sim t_n$. The t_1 -distribution is also called Cauchy distribution.

If $X \sim N(0, 1)$ and $Y \sim \chi_n^2$ are independent, then

$$T = \frac{X}{\sqrt{\frac{Y}{n}}} \sim t_n. \quad (\text{B.1})$$

If $X_1, \dots, X_n$ are i.i.d. $N(\mu, \sigma^2)$ random variables, then

$$\frac{\bar{X} - \mu}{S} \sqrt{n} \sim t_{n-1}$$

with

$$S = \frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X})^2 \quad \text{and} \quad \bar{X} = \frac{1}{n} \sum_{i=1}^n X_i.$$

Definition B.14 F-Distribution

A continuous random variable X is said to have a F-distribution with n and m degrees of freedom if it has p.d.f.

$$f(x) = n^{n/2} m^{m/2} \frac{\Gamma(n/2 + m/2)}{\Gamma(n/2) \Gamma(m/2)} \frac{x^{n/2-1}}{(nx + m)^{(n+m)/2}}, \quad x \geq 0.$$

We write $F \sim F_{n,m}$.

If $X_1 \sim \chi_n^2$ and $X_2 \sim \chi_m^2$ are independent, then

$$X = \frac{X_1/n}{X_2/m}$$

has an F-distribution with n and m degrees of freedom.

If Y is t -distributed with m degrees of freedom then $X = Y^2 \sim F_{1,m}$.

Theorem B.1 Change of Variables

Let X be a continuous random variable with p.d.f. $f_X(x)$ and $Y = g(X)$ a monotonic transformation of X . It follows that Y has p.d.f.

$$f_Y(y) = \frac{f_X(g^{-1}(y))}{|g'(g^{-1}(y))|}.$$

B.2 Random Vectors**Definition B.15 Random Vector**

The p -dimensional vector $\mathbf{X} = (X_1, \dots, X_p)'$ is called random vector or p -dimensional random variable if the components $X_1, \dots, X_p$ are univariate (scalar) random variables. The random vector $\mathbf{X}$ is called continuous if there is a function $f(\mathbf{x}) = f(x_1, \dots, x_p) \geq 0$ such that

$$P(a_1 \leq X_1 \leq b_1, \dots, a_p \leq X_p \leq b_p) = \int_{a_p}^{b_p} \dots \int_{a_1}^{b_1} f(x_1, \dots, x_p) dx_1 \dots dx_p.$$

The function f is called (joint) probability density function (p.d.f.) of $\mathbf{X}$.

The random vector $\mathbf{X}$ is called discrete, if $\mathbf{X}$ has only values in a finite or countable set $\{\mathbf{x}_1, \mathbf{x}_2, \dots\} \subset \mathbb{R}^p$. The function f with

$$f(\mathbf{x}) = \begin{cases} P(\mathbf{X} = \mathbf{x}) & \mathbf{x} \in \{\mathbf{x}_1, \mathbf{x}_2, \dots\} \\ 0 & \text{else} \end{cases}$$

is called probability function or discrete p.d.f. of $\mathbf{X}$.

Definition B.16 Marginal and Conditional Distributions

Let the p -dimensional random vector $X = (X_1, \dots, X_p)'$ be partitioned into the p_1 -dimensional vector X_1 and the p_2 -dimensional vector X_2 , i.e., $X = (X_1', X_2')'$. The p_1 -dimensional p.d.f. or probability function $f_{X_1}(x_1)$ of X_1 is then called marginal p.d.f. or marginal probability function of X . It is given by

$$f_{X_1}(x_1) = \int_{-\infty}^{\infty} \dots \int_{-\infty}^{\infty} f(x_1, x_2) dx_{p_1+1} \dots dx_p$$

for continuous random vectors and

$$f_{X_1}(x_1) = \sum_{x_2} f(x_1, x_2)$$

for discrete random vectors. The conditional p.d.f. or probability function of X_1 given $X_2 = x_2$ is defined as

$$f(x_1|x_2) = \begin{cases} \frac{f(x_1, x_2)}{f_{X_2}(x_2)} & \text{for } f_{X_2}(x_2) > 0 \\ 0 & \text{else.} \end{cases}$$

The marginal and conditional p.d.f.'s or probability functions for X_2 are defined in complete analogy.

Definition B.17 Mean Vector

Let $X = (X_1, \dots, X_p)'$ be a p -dimensional random vector. Then

$$E(X) = \mu = (\mu_1, \dots, \mu_p)' = (E(X_1), \dots, E(X_p))'$$

is called mean vector of X .

Definition B.18 Covariance Matrix, Correlation Matrix, and Precision Matrix

The covariance matrix $\text{Cov}(X) = \Sigma$ of a p -dimensional random vector X is defined as

$$\text{Cov}(X) = \Sigma = E(X - \mu)(X - \mu)' = \begin{pmatrix} \sigma_{11} & \dots & \sigma_{1p} \\ \vdots & & \vdots \\ \sigma_{p1} & \dots & \sigma_{pp} \end{pmatrix},$$

where $\sigma_{ij} = \text{Cov}(X_i, X_j)$, $i \neq j$, is the covariance between X_i and X_j , and $\sigma_{ii} = \sigma_i^2 = \text{Var}(X_i)$ is the variance of X_i .

The correlation matrix $\mathbf{R}$ of $\mathbf{X}$ is defined as

$$\mathbf{R} = \begin{pmatrix} 1 & \rho_{12} & \dots & \rho_{1p} \\ \vdots & & & \vdots \\ \rho_{p1} & \rho_{p2} & \dots & 1 \end{pmatrix},$$

where

$$\rho_{ij} = \frac{\text{Cov}(X_i, X_j)}{\sqrt{\text{Var}(X_i) \cdot \text{Var}(X_j)}}.$$

The covariance matrix $\mathbf{\Sigma}$ as well as the correlation matrix $\mathbf{R}$ are symmetric and positive semidefinite. They are positive definite if the components $X_1, \dots, X_p$ are linearly independent, i.e., no component can be expressed as a linear combination of remaining components. If $\mathbf{\Sigma}$ is positive definite, then the inverse $\mathbf{P} = \mathbf{\Sigma}^{-1}$ is called precision matrix. In case that $\mathbf{\Sigma}$ is only positive semidefinite and therefore not invertible, the precision matrix is a generalized inverse, i.e., $\mathbf{P} = \mathbf{\Sigma}^-$.

Theorem B.2 Rules for Mean Vectors and Covariance Matrices

Let $\mathbf{X}$, $\mathbf{Y}$, and $\mathbf{Z}$ be random vectors; $\mathbf{A}$, $\mathbf{B}$, $\mathbf{a}$, $\mathbf{b}$ matrices or vectors of appropriate order; and $\mathbf{E}(\mathbf{X}) = \boldsymbol{\mu}$, $\text{Cov}(\mathbf{X}) = \mathbf{\Sigma}$. The following rules then hold:

1. $\mathbf{E}(\mathbf{X} + \mathbf{Y}) = \mathbf{E}(\mathbf{X}) + \mathbf{E}(\mathbf{Y})$
2. $\mathbf{E}(\mathbf{A}\mathbf{X} + \mathbf{b}) = \mathbf{A} \cdot \mathbf{E}(\mathbf{X}) + \mathbf{b}$
3. $\text{Cov}(\mathbf{X}) = \mathbf{E}(\mathbf{X}\mathbf{X}') - \mathbf{E}(\mathbf{X})\mathbf{E}(\mathbf{X})'$
4. $\text{Var}(\mathbf{a}'\mathbf{X}) = \mathbf{a}'\text{Cov}(\mathbf{X})\mathbf{a} = \sum_{i=1}^p \sum_{j=1}^p a_i a_j \sigma_{ij}$
5. $\text{Cov}(\mathbf{A}\mathbf{X} + \mathbf{b}) = \mathbf{A}\text{Cov}(\mathbf{X})\mathbf{A}'$
6. $\text{Cov}(\mathbf{X} + \mathbf{Y}) = \text{Cov}(\mathbf{X}) + \text{Cov}(\mathbf{Y})$ provided that $\mathbf{X}$ and $\mathbf{Y}$ are uncorrelated
7. $\text{Cov}(\mathbf{X} + \mathbf{Y}, \mathbf{Z}) = \text{Cov}(\mathbf{X}, \mathbf{Z}) + \text{Cov}(\mathbf{Y}, \mathbf{Z})$
8. $\mathbf{E}(\mathbf{X}'\mathbf{A}\mathbf{X}) = \text{tr}(\mathbf{A}\mathbf{\Sigma}) + \boldsymbol{\mu}'\mathbf{A}\boldsymbol{\mu}$
9. Law of total (or iterated) expectation: $\mathbf{E}(\mathbf{X}) = \mathbf{E}(\mathbf{E}(\mathbf{X} | \mathbf{Y}))$
10. Law of total covariance: $\text{Cov}(\mathbf{X}) = \mathbf{E}(\text{Cov}(\mathbf{X} | \mathbf{Y})) + \text{Cov}(\mathbf{E}(\mathbf{X} | \mathbf{Y}))$
11. Law of total variance for scalar random variables X : $\text{Var}(X) = \mathbf{E}(\text{Var}(X | \mathbf{Y})) + \text{Var}(\mathbf{E}(X | \mathbf{Y}))$

Definition B.19 Empirical Means, Covariance, and Correlation Matrices

Let $\mathbf{x}_1, \dots, \mathbf{x}_n$ be realizations of i.i.d. random vectors $\mathbf{X}_1, \dots, \mathbf{X}_n$ from the distribution of $\mathbf{X}$. The mean vector then can be estimated by the empirical mean

$$\hat{\boldsymbol{\mu}} = (\hat{\mu}_1, \dots, \hat{\mu}_p)' = (\bar{x}_1, \dots, \bar{x}_p)' = \bar{\mathbf{x}},$$

where

$$\hat{\mu}_j = \frac{1}{n} \sum_{i=1}^n x_{ij} = \bar{x}_j.$$

Correspondingly, the empirical covariance matrix $\hat{\Sigma}$ is defined by estimating variances and covariances in Σ through the empirical variances

$$\hat{\sigma}_j^2 = \frac{1}{n-1} \sum_{i=1}^n (x_{ij} - \bar{x}_j)^2$$

and the empirical covariances

$$\hat{\sigma}_{jk} = \frac{1}{n-1} \sum_{i=1}^n (x_{ij} - \bar{x}_j)(x_{ik} - \bar{x}_k).$$

The empirical correlation matrix $\hat{R}$ is defined analogously through the empirical correlation coefficients.

B.3 Multivariate Normal Distribution

B.3.1 Definition and Properties

Definition B.20 Multivariate Normal Distribution

A continuous p -dimensional random vector $X = (X_1, X_2, \dots, X_p)'$ is said to have a multivariate normal (or Gaussian) distribution if it has p.d.f.

$$f(x) = (2\pi)^{-\frac{p}{2}} |\Sigma|^{-\frac{1}{2}} \exp \left[-\frac{1}{2} (x - \mu)' \Sigma^{-1} (x - \mu) \right] \quad (\text{B.2})$$

with $\mu \in \mathbb{R}^p$ and positive definite $(p \times p)$ -matrix Σ .

For deriving posterior distributions in Bayesian models it is often convenient to rewrite the p.d.f. in alternative form. Omitting all factors in Eq. (B.2) that do not depend on x , we obtain

$$\begin{aligned} f(x) &\propto \exp \left(-\frac{1}{2} (x - \mu)' \Sigma^{-1} (x - \mu) \right) \\ &= \exp \left(-\frac{1}{2} x' \Sigma^{-1} x + x' \Sigma^{-1} \mu - \frac{1}{2} \mu' \Sigma^{-1} \mu \right) \\ &\propto \exp \left(-\frac{1}{2} x' \Sigma^{-1} x + x' \Sigma^{-1} \mu \right). \end{aligned} \quad (\text{B.3})$$

If X is multivariate normal, then its p.d.f. is always proportional to Eq. (B.3).

Theorem B.3 Mean and Covariance Matrix

It can be shown that $E(X) = \mu$ and $\text{Cov}(X) = \Sigma$. We write

$$X \sim N_p(\mu, \Sigma),$$

analogous to the univariate normal distribution. The index p is often suppressed if the dimension p is obvious from the context. The special case $\boldsymbol{\mu} = \mathbf{0}$ and $\boldsymbol{\Sigma} = \mathbf{I}$ is called the (multivariate) standard normal distribution.

Theorem B.4 Product of Two Normal Densities

Suppose that the density of a random vector $\mathbf{X}$ is given by

$$f(\mathbf{x}) \propto \exp\left(-\frac{1}{2}(\mathbf{x} - \mathbf{a})' \mathbf{A}(\mathbf{x} - \mathbf{a})\right) \exp\left(-\frac{1}{2}(\mathbf{x} - \mathbf{b})' \mathbf{B}(\mathbf{x} - \mathbf{b})\right).$$

$\mathbf{X}$ is then multivariate normal with covariance matrix and mean

$$\boldsymbol{\Sigma} = (\mathbf{A} + \mathbf{B})^{-1} \quad \boldsymbol{\mu} = \boldsymbol{\Sigma}(\mathbf{A}\mathbf{a} + \mathbf{B}\mathbf{b}).$$

Theorem B.5 Linear Transformation

Let $\mathbf{X} \sim N_p(\boldsymbol{\mu}, \boldsymbol{\Sigma})$, $\mathbf{d} \in \mathbb{R}^q$, and $\mathbf{D}$ a $(q \times p)$ -matrix with $\text{rk}(\mathbf{D}) = q \leq p$; then

$$\mathbf{Y} = \mathbf{d} + \mathbf{D}\mathbf{X} \sim N_q(\mathbf{d} + \mathbf{D}\boldsymbol{\mu}, \mathbf{D}\boldsymbol{\Sigma}\mathbf{D}').$$

Theorem B.6 Marginal and Conditional Distributions

Let the multivariate normal random variable $\mathbf{X} \sim N(\boldsymbol{\mu}, \boldsymbol{\Sigma})$ be partitioned into the subvectors $\mathbf{Y} = (X_1, \dots, X_r)'$ and $\mathbf{Z} = (X_{r+1}, \dots, X_p)'$, i.e.,

$$\mathbf{X} = \begin{pmatrix} \mathbf{Y} \\ \mathbf{Z} \end{pmatrix}, \quad \boldsymbol{\mu} = \begin{pmatrix} \boldsymbol{\mu}_Y \\ \boldsymbol{\mu}_Z \end{pmatrix}, \quad \boldsymbol{\Sigma} = \begin{pmatrix} \boldsymbol{\Sigma}_Y & \boldsymbol{\Sigma}_{YZ} \\ \boldsymbol{\Sigma}_{ZY} & \boldsymbol{\Sigma}_Z \end{pmatrix}.$$

Then $\mathbf{Y}$ has an r -dimensional normal distribution $\mathbf{Y} \sim N(\boldsymbol{\mu}_Y, \boldsymbol{\Sigma}_Y)$.

The conditional distribution of $\mathbf{Y}$ given $\mathbf{Z}$ is again multivariate normal with mean

$$\boldsymbol{\mu}_{Y|Z} = \boldsymbol{\mu}_Y + \boldsymbol{\Sigma}_{YZ} \cdot \boldsymbol{\Sigma}_Z^{-1}(\mathbf{Z} - \boldsymbol{\mu}_Z)$$

and covariance matrix

$$\boldsymbol{\Sigma}_{Y|Z} = \boldsymbol{\Sigma}_Y - \boldsymbol{\Sigma}_{YZ} \boldsymbol{\Sigma}_Z^{-1} \boldsymbol{\Sigma}_{ZY}.$$

Furthermore, $\mathbf{Y}$ and $\mathbf{Z}$ are independent if and only if $\mathbf{Y}$ and $\mathbf{Z}$ are uncorrelated, i.e., if $\boldsymbol{\Sigma}_{YZ} = \boldsymbol{\Sigma}_{ZY} = \mathbf{0}$. This equivalence is generally not true for non-normal random vectors: If $\mathbf{Y}$ and $\mathbf{Z}$ are independent they are also uncorrelated, but in general $\boldsymbol{\Sigma}_{ZY} = \mathbf{0}$ does not imply independence.

B.3.2 The Singular Multivariate Normal Distribution

Up to now, we have assumed that $\text{rk}(\Sigma) = p$ so that Σ is positive definite and Σ^{-1} exists. In the following, we consider the case $\text{rk}(\Sigma) < p$.

Definition B.21 Singular Multivariate Normal Distribution

Let $X \sim N_p(\mu, \Sigma)$. Then X follows a singular multivariate normal distribution if $\text{rk}(\Sigma) = r < p$. The distribution of a singular normal distribution is often expressed in terms of the precision matrix P with $\text{rk}(P) = r < p$. The density is then given by

$$f(x) \propto \exp \left[-\frac{1}{2}(x - \mu)' P (x - \mu) \right].$$

The precision matrix P may be chosen as a generalized inverse Σ^- of Σ . This form of the singular normal distribution will be used frequently in the book, in particular in Chaps. 8 and 9.

The virtue of specifying in statistical models the precision matrix rather than the covariance matrix is that the elements of the precision matrix can be interpreted as conditional correlations. More specifically, for multivariate normal distributions we have

$$\text{Corr}(X_i, X_j \mid X_{-ij}) = -\frac{p_{ij}}{\sqrt{p_{ii} p_{jj}}},$$

where X_{-ij} is the vector X with the i th and j th element excluded and p_{ij} is the element of P in the i th row and j th column. Thus, $p_{ij} = 0$ in the precision matrix implies that X_i and X_j are conditionally uncorrelated and (in case of the normal distribution) also conditionally independent. Note that conditional independence does not imply (unconditional) independence.

Theorem B.7 Characterization of the Singular Normal Distribution

Let X be a singular normal random vector, i.e., $X \sim N_p(\mu, \Sigma)$, with $\text{rk}(\Sigma) = \text{rk}(P) = r < p$. Assume that $(G \ H)$ is an orthogonal matrix, where the columns of the $(p \times r)$ -matrix G are a basis for the column space of Σ and the columns of H are a basis of the null space of Σ . Consider the transformation

$$\begin{pmatrix} Y_1 \\ Y_2 \end{pmatrix} = (G \ H)' X = \begin{pmatrix} G' X \\ H' X \end{pmatrix}.$$

It follows that Y_1 is the stochastic part of X and is nonsingular normal with

$$Y_1 \sim N(G' \mu, G' \Sigma G).$$

Y_2 is the deterministic part of X with

$$E(Y_2) = H' \mu, \quad \text{Var}(Y_2) = 0.$$

Here, $\text{Var}(\mathbf{Y}_2)$ is the vector of variances of $\mathbf{Y}_2$. The p.d.f. of the stochastic part $\mathbf{Y}_1 = \mathbf{G}'\mathbf{X}$ has the form

$$f(\mathbf{y}_1) = \frac{1}{(2\pi)^{\frac{r}{2}} (\prod_{i=1}^r \lambda_i)^{\frac{1}{2}}} \exp \left[-\frac{1}{2} (\mathbf{y}_1 - \mathbf{G}'\boldsymbol{\mu})' (\mathbf{G}'\boldsymbol{\Sigma}\mathbf{G})^{-1} (\mathbf{y}_1 - \mathbf{G}'\boldsymbol{\mu}) \right], \quad (\text{B.4})$$

where $\lambda_1, \dots, \lambda_r$ are the r nonzero eigenvalues of $\boldsymbol{\Sigma}$.

B.3.3 Distributions of Quadratic Forms

Quadratic forms of normal random vectors appear in tests of linear hypotheses. Refer to Sect. 3.3 in Chap. 3.

Theorem B.8 Distributions of Quadratic Forms

1. Let $\mathbf{X} \sim N_p(\boldsymbol{\mu}, \boldsymbol{\Sigma})$ with $\boldsymbol{\Sigma} > \mathbf{0}$. It follows that

$$\mathbf{Y} = (\mathbf{X} - \boldsymbol{\mu})' \boldsymbol{\Sigma}^{-1} (\mathbf{X} - \boldsymbol{\mu}) \sim \chi_p^2.$$

2. Let $\mathbf{X} \sim N_p(\mathbf{0}, \mathbf{I})$, $\mathbf{B}$ an $(n \times p)$ -matrix ($n \leq p$) and, $\mathbf{R}$ a symmetric idempotent $(p \times p)$ -matrix with $\text{rk}(\mathbf{R}) = r$; then
 - $\mathbf{X}'\mathbf{R}\mathbf{X} \sim \chi_r^2$.
 - $\mathbf{B}\mathbf{R} = \mathbf{0}$ implies independence of $\mathbf{X}'\mathbf{R}\mathbf{X}$ and $\mathbf{B}\mathbf{X}$.
3. Assume that $X_1, \dots, X_n$ are i.i.d. $\sim N(\mu, \sigma^2)$ variables and

$$S^2 = \frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X})^2.$$

It then follows that:

- $\frac{n-1}{\sigma^2} S^2 \sim \chi_{n-1}^2$.
 - S^2 and $\bar{X}$ are independent.
4. Assume that $\mathbf{X} \sim N_n(\mathbf{0}, \mathbf{I})$, and let $\mathbf{R}$ and $\mathbf{S}$ be symmetric and idempotent $n \times n$ matrices with $\text{rk}(\mathbf{R}) = r$, $\text{rk}(\mathbf{S}) = s$, and $\mathbf{R}\mathbf{S} = \mathbf{0}$. It follows that
 - $\mathbf{X}'\mathbf{R}\mathbf{X}$ and $\mathbf{X}'\mathbf{S}\mathbf{X}$ are independent.
 - $\frac{s}{r} \frac{\mathbf{X}'\mathbf{R}\mathbf{X}}{\mathbf{X}'\mathbf{S}\mathbf{X}} \sim F_{r,s}$.

B.3.4 Multivariate t-Distribution

A continuous p -dimensional random vector $\mathbf{X} = (X_1, \dots, X_p)'$ is said to have a multivariate t-distribution with ν degrees of freedom, location parameter $\boldsymbol{\mu}$, and (positive definite) dispersion matrix $\boldsymbol{\Sigma}$, if it has p.d.f.

$$f(\mathbf{x}) = |\boldsymbol{\Sigma}|^{-\frac{1}{2}} (\nu\pi)^{-\frac{p}{2}} \frac{\Gamma((\nu + p)/2)}{\Gamma(\nu/2)} \left(1 + \frac{(\mathbf{x} - \boldsymbol{\mu})' \boldsymbol{\Sigma}^{-1} (\mathbf{x} - \boldsymbol{\mu})}{\nu} \right)^{-(\nu+p)/2}.$$

We write $\mathbf{X} \sim t(\nu, \boldsymbol{\mu}, \boldsymbol{\Sigma})$. The expectation is $\boldsymbol{\mu}$ (provided that $\nu > 1$) and the covariance matrix is $\nu/(\nu - 2) \boldsymbol{\Sigma}$ (provided that $\nu > 2$). Note that a diagonal dispersion matrix $\boldsymbol{\Sigma}$ corresponds to uncorrelated components of the random vector $\mathbf{X}$. In contrast to the multivariate normal distribution the components are, however, not stochastically independent.

Any subvector of $\mathbf{X}$ has a (multivariate) t-distribution with ν degrees of freedom and the corresponding subvector of $\boldsymbol{\mu}$ and the submatrix of $\boldsymbol{\Sigma}$ as location parameter and dispersion matrix, respectively.

In analogy to the constructive definition (B.1) of the univariate t-distribution, the multivariate t-distribution can be defined constructively, based on a multivariate normal random vector and a chi-squared distributed random variable.

B.3.5 Normal-Inverse Gamma Distribution

Let $\mathbf{Y}$ be a $p \times 1$ dimensional random vector and S be a random variable. The random vector $\mathbf{X} = (\mathbf{Y}, S)'$ is said to have a normal-inverse gamma distribution with parameters $\boldsymbol{\mu}$, $\boldsymbol{\Sigma}$, a , and b if

$$\mathbf{Y} | S \sim N(\boldsymbol{\mu}, S \boldsymbol{\Sigma}),$$

$$S \sim \text{IG}(a, b).$$

We write $\mathbf{X} = (\mathbf{Y}, S)' \sim \text{NIG}(\boldsymbol{\mu}, \boldsymbol{\Sigma}, a, b)$. The density of the distribution is given by

$$f(\mathbf{y}, s) = \frac{1}{(2\pi)^{\frac{p}{2}} |\boldsymbol{\Sigma}|^{\frac{1}{2}}} \exp\left(-\frac{1}{2}(\mathbf{y} - \boldsymbol{\mu})' \boldsymbol{\Sigma}^{-1} (\mathbf{y} - \boldsymbol{\mu})\right) \frac{b^a}{\Gamma(a)} \frac{1}{(\sigma^2)^{a+1}} \exp\left(-\frac{b}{\sigma^2}\right).$$

The $\text{NIG}(\boldsymbol{\mu}, \boldsymbol{\Sigma}, a, b)$ -distribution has the following properties:

1. $E(\mathbf{Y}) = \boldsymbol{\mu}$
2. $\text{Cov}(\mathbf{Y}) = b/(a - 1) \boldsymbol{\Sigma}$
3. $E(S) = b/(a - 1)$ provided that $a > 1$
4. $\text{Var}(S) = b^2/[(a - 1)^2(a - 2)]$ provided that $a > 2$
5. $\mathbf{Y} \sim t(2a, \boldsymbol{\mu}, b/a \boldsymbol{\Sigma})$

B.4 Likelihood Inference

This section describes the method of maximum likelihood (ML) for estimation of unknown parameters in statistical models and likelihood-based tests for linear hypotheses about these parameters. More detailed introductions can be found, for example, in [Migon and Gamerman \(1999\)](#) and [Held and Sabanés Bové \(2012\)](#).

B.4.1 Maximum Likelihood Estimation

Let $Y_1, \dots, Y_n$ be a random sample with realizations $y_1, \dots, y_n$. Let the joint probability (for discrete $Y_1, \dots, Y_n$)

$$P(Y_1 = y_1, \dots, Y_n = y_n | \theta)$$

or the joint p.d.f. (for continuous $Y_1, \dots, Y_n$)

$$f(Y_1 = y_1, \dots, Y_n = y_n | \theta)$$

depend on an unknown vector $\theta = (\theta_1, \dots, \theta_p)' \in \Theta$ that has to be estimated. For given realizations $y_1, \dots, y_n$ this joint probability or the value of the p.d.f. is considered as a function of θ and is called likelihood, denoted by $L(\theta)$:

$$L(\theta) = P(Y_1 = y_1, \dots, Y_n = y_n | \theta),$$

$$L(\theta) = f(Y_1 = y_1, \dots, Y_n = y_n | \theta).$$

The *principle of maximum likelihood* postulates the *maximum likelihood estimator* (MLE) $\hat{\theta}$ for θ as the value $\hat{\theta}$ which maximizes the likelihood $L(\theta)$.

In the discrete case, this principle says: Determine the MLE $\hat{\theta}$ such that the probability of observing the realized sample $y_1, \dots, y_n$ assumes its maximum for $\theta = \hat{\theta}$ and, therefore, this sample becomes as likely as possible. In analogy, the p.d.f. assumes its maximum for $\theta = \hat{\theta}$, making the sample as plausible as possible. In most cases, in particular for almost all models in this book (excluding quantile regression), the likelihood $L(\theta)$ is differentiable with respect to θ , and the maximum can be determined by setting the first derivatives to zero and solving the resulting system of equations for $\theta = \hat{\theta}$. For technical reasons, maximization is usually not carried out for the likelihood but for the log-likelihood $l(\theta) = \log(L(\theta))$, obtained by taking the logarithm of the likelihood. Since the logarithm is a strictly increasing function, $l(\theta)$ attains its maximum at the same value $\theta = \hat{\theta}$ as $L(\theta)$.

In the most simple situation, $Y_1, \dots, Y_n$ are an i.i.d. sample from $f(y | \theta)$, i.e., they are independent and identically distributed as the typical random variable $Y \sim f(y | \theta)$. To unify notation, we denote probability functions (Y discrete) and p.d.f.'s (Y continuous) with $f(y | \theta)$. Because $Y_1, \dots, Y_n$ are independent, their joint density

is the product of the densities $f(y_i | \theta)$. Therefore the likelihood is

$$L(\theta) = f(y_1 | \theta) \cdot \dots \cdot f(y_n | \theta)$$

and the log-likelihood is the sum

$$l(\theta) = \log f(y_1 | \theta) + \dots + \log f(y_n | \theta) = \sum_{i=1}^n l_i(\theta)$$

with the log-likelihood contributions $l_i(\theta) = \log f(y_i | \theta)$. The following example illustrates this situation.

Example B.1 Poisson Distribution: MLE

Let $Y \sim \text{Po}(\lambda)$ follow a Poisson distribution with unknown parameter λ . To estimate λ , we assume an i.i.d. sample $Y_1, \dots, Y_n$ with $Y_i \sim \text{Po}(\lambda)$ and a realized sample $y_1, \dots, y_n$. The MLE for λ is obtained in the following steps:

1. Likelihood

Because $Y_1, \dots, Y_n$ are independent, the joint probability for the realized sample factorizes into the product of the separate marginal probabilities, and we obtain the likelihood

$$\begin{aligned} L(\lambda) &= P(Y_1 = y_1, \dots, Y_n = y_n | \lambda) \\ &= P(Y_1 = y_1 | \lambda) \cdot \dots \cdot P(Y_n = y_n | \lambda) \\ &= \frac{\lambda^{y_1}}{y_1!} \exp(-\lambda) \cdot \dots \cdot \frac{\lambda^{y_n}}{y_n!} \exp(-\lambda). \end{aligned}$$

The factors $1/y_1!, \dots, 1/y_n!$ do not depend on λ and can therefore be omitted when maximizing the likelihood. Therefore

$$L(\lambda) \propto \exp(-n\lambda) \cdot \lambda^{y_1} \cdot \dots \cdot \lambda^{y_n}.$$

Often the right-hand side, after omitting constant factors, is also called likelihood, and we write

$$L(\lambda) = \exp(-n\lambda) \cdot \lambda^{y_1} \cdot \dots \cdot \lambda^{y_n}.$$

2. Log-likelihood

Taking logarithms, we get

$$l(\lambda) = -n\lambda + \sum_{i=1}^n y_i \log(\lambda).$$

3. First derivative and setting it to zero

$$\frac{\partial l(\lambda)}{\partial \lambda} = -n + \sum_{i=1}^n y_i \frac{1}{\lambda} \stackrel{!}{=} 0.$$

The first derivative $s(\lambda) = \partial l(\lambda) / \partial \lambda$ is called *score function*.

Solving for the unknown parameter gives the MLE:

$$\hat{\lambda} = \frac{1}{n} \sum_{i=1}^n y_i = \bar{y}.$$

4. *Second derivative to check for a maximum*

The second derivative of the log-likelihood is

$$\frac{\partial^2 l(\lambda)}{\partial \lambda^2} = - \sum_{i=1}^n y_i \frac{1}{\lambda^2} < 0.$$

Therefore the estimator of step 3 is indeed a maximizer of the log-likelihood (and the likelihood). The negative second derivative is called *observed Fisher information*. $\triangle$

The score function and the Fisher information appearing in this example play an important role in likelihood theory. Generally, the score function is defined as the vector

$$s(\boldsymbol{\theta}) = (s_1(\boldsymbol{\theta}), \dots, s_p(\boldsymbol{\theta}))' = \left(\frac{\partial l(\boldsymbol{\theta})}{\partial \theta_1}, \dots, \frac{\partial l(\boldsymbol{\theta})}{\partial \theta_p} \right)' = \frac{\partial l(\boldsymbol{\theta})}{\partial \boldsymbol{\theta}}$$

of partial first derivatives of the log-likelihood. The observed Fisher information (matrix) is defined as

$$H(\boldsymbol{\theta}) = - \begin{pmatrix} \frac{\partial^2 l(\boldsymbol{\theta})}{\partial \theta_1 \partial \theta_1} & \dots & \frac{\partial^2 l(\boldsymbol{\theta})}{\partial \theta_1 \partial \theta_p} \\ \vdots & & \vdots \\ \frac{\partial^2 l(\boldsymbol{\theta})}{\partial \theta_p \partial \theta_1} & \dots & \frac{\partial^2 l(\boldsymbol{\theta})}{\partial \theta_p \partial \theta_p} \end{pmatrix} = - \begin{pmatrix} \frac{\partial s_1(\boldsymbol{\theta})}{\partial \theta_1} & \dots & \frac{\partial s_1(\boldsymbol{\theta})}{\partial \theta_p} \\ \vdots & & \vdots \\ \frac{\partial s_p(\boldsymbol{\theta})}{\partial \theta_1} & \dots & \frac{\partial s_p(\boldsymbol{\theta})}{\partial \theta_p} \end{pmatrix},$$

i.e., the negative matrix of second derivatives of the log-likelihood. In more compact notation we obtain

$$H(\boldsymbol{\theta}) = - \left(\frac{\partial s(\boldsymbol{\theta})}{\partial \boldsymbol{\theta}} \right)' = - \frac{\partial s(\boldsymbol{\theta})}{\partial \boldsymbol{\theta}'} = - \frac{\partial^2 l(\boldsymbol{\theta})}{\partial \boldsymbol{\theta} \partial \boldsymbol{\theta}'},$$

thereby using Definition A.29 (p. 636) in Appendix A.8. As the name already indicates, the observed Fisher information matrix can be considered as a (local) measure of the information that the likelihood contains about the unknown parameter. The second derivative of a function is a measure of the curvature of the function at $\boldsymbol{\theta}$. The higher the curvature of the log-likelihood near its maximum, the more information is provided by the likelihood about the unknown parameter. Since the second derivative is negative at or near a maximum, the Fisher information is defined as the negative second derivative, i.e., it is positive at or near a maximum.

Since the likelihood (as well as the log-likelihood) depends on the realized values of the sample variables $Y_1, \dots, Y_n$, the likelihood function will have different values when samples are repeatedly drawn from the (joint) distribution of the sample.

Therefore, the likelihood as well as the log-likelihood and its derivatives can be interpreted as random variables, having mean, variance, etc. Under mild regularity conditions that are fulfilled for all examples and models in this book, the score function has mean $\mathbf{0}$, i.e.,

$$E(s(\theta)) = \mathbf{0}.$$

The mean is taken with respect to the sample variables $Y_1, \dots, Y_n$. The mean of the observed Fisher information (matrix)

$$F(\theta) = - \begin{pmatrix} E\left(\frac{\partial^2 l(\theta)}{\partial \theta_1 \partial \theta_1}\right) & \dots & E\left(\frac{\partial^2 l(\theta)}{\partial \theta_1 \partial \theta_p}\right) \\ \vdots & & \vdots \\ E\left(\frac{\partial^2 l(\theta)}{\partial \theta_p \partial \theta_1}\right) & \dots & E\left(\frac{\partial^2 l(\theta)}{\partial \theta_p \partial \theta_p}\right) \end{pmatrix} = - \begin{pmatrix} E\left(\frac{\partial s_1(\theta)}{\partial \theta_1}\right) & \dots & E\left(\frac{\partial s_1(\theta)}{\partial \theta_p}\right) \\ \vdots & & \vdots \\ E\left(\frac{\partial s_p(\theta)}{\partial \theta_1}\right) & \dots & E\left(\frac{\partial s_p(\theta)}{\partial \theta_p}\right) \end{pmatrix}$$

is of particular interest and is called expected Fisher information. It can be considered as a global measure of information which can be determined prior to and independent from realized samples. Under mild regularity conditions fulfilled for all models throughout this book it can be shown that

$$F(\theta) = \text{Cov}(s(\theta)) = E(s(\theta)s(\theta)').$$

The latter equality holds because $E(s(\theta)) = \mathbf{0}$.

For independent sample variables information is additive: Let $H_{Y_i}(\theta)$ and $F_{Y_i}(\theta)$ denote the observed and expected Fisher information with respect to the sample variable Y_i , i.e., the information contributed by Y_i alone; then the complete information provided by the sample is

$$H_Y(\theta) = \sum_{i=1}^n H_{Y_i}(\theta) \quad , \quad F_Y(\theta) = \sum_{i=1}^n F_{Y_i}(\theta).$$

Similarly, the score function is additive and composed of the individual score contributions:

$$s(\theta) = \sum_{i=1}^n s_i(\theta).$$

Example B.2 Poisson Distribution: Score Function and Fisher Information

Let $Y_1, \dots, Y_n$ be an i.i.d. sample for a Poisson variable $Y \sim \text{Po}(\lambda)$ with unknown parameter λ . The log-likelihood and score function contribution of y_i are (refer to Example B.1)

$$l_i(\lambda) = y_i \log(\lambda) - \log(y_i!) - \lambda \quad \text{and} \quad s_i(\lambda) = \frac{y_i}{\lambda} - 1.$$

It is easy to see that $E(s(\lambda)) = 0$: Since

$$E(s_i(\lambda)) = E\left(\frac{Y_i}{\lambda} - 1\right) = \frac{1}{\lambda}E(Y_i) - 1 = \frac{1}{\lambda}\lambda - 1 = 0,$$

we have

$$E(s(\lambda)) = E\left(\sum_{i=1}^n s_i(\lambda)\right) = \sum_{i=1}^n E(s_i(\lambda)) = 0.$$

The observed Fisher information contributed by y_i is

$$H_{y_i}(\lambda) = -\frac{\partial^2 l_i(\lambda)}{\partial^2 \lambda} = -\frac{\partial s_i(\lambda)}{\partial \lambda} = -\left(-\frac{y_i}{\lambda^2}\right) = \frac{y_i}{\lambda^2}.$$

Due to additivity of information, we obtain the observed information

$$H_Y(\lambda) = \frac{1}{\lambda^2} \sum_{i=1}^n y_i$$

for the entire sample. The expected Fisher information contributed by Y_i , with $E(Y_i) = \lambda$, is

$$F_{Y_i}(\lambda) = E\left(\frac{Y_i}{\lambda^2}\right) = \frac{\lambda}{\lambda^2} = \frac{1}{\lambda},$$

and we can verify additivity, i.e.,

$$F_Y(\lambda) = \frac{n}{\lambda}$$

for the entire sample. This shows that expected information grows linearly with sample size n . Furthermore, it is inversely proportional to λ which is plausible because the variance of the MLE $\bar{Y}$ is $\text{Var}(\bar{Y}) = \frac{1}{n}\lambda$, i.e., the inverse of the expected information increases linearly with λ . $\triangle$

In this book, we mostly consider regression situations with response variables $Y_1, \dots, Y_n$ and with observed responses $y_1, \dots, y_n$, as well as additional covariate values $\mathbf{x}_1, \dots, \mathbf{x}_n$ as the realized sample. The vector $\boldsymbol{\theta}$ is then often the vector $\boldsymbol{\beta}$ of regression coefficients. Given $\mathbf{x}_1, \dots, \mathbf{x}_n$, the response variables $Y_1, \dots, Y_n$ are no longer identically distributed, but they are still assumed to be conditionally independent given the covariates. The densities of the Y_i depend on $\mathbf{x}_i$ and, therefore, on i , i.e., $Y_i \sim f_i(y_i | \boldsymbol{\theta}) = f(y_i | \mathbf{x}_i; \boldsymbol{\beta})$ with $\boldsymbol{\theta} = \boldsymbol{\beta}$. Due to independence of the sample variables, the likelihood is again the product

$$L(\boldsymbol{\theta}) = f_1(y_1 | \boldsymbol{\theta}) \cdots \cdots f_n(y_n | \boldsymbol{\theta})$$

of individual densities, and the log-likelihood is the sum

$$l(\boldsymbol{\theta}) = \sum_{i=1}^n l_i(\boldsymbol{\theta}) = \sum_{i=1}^n \log f_i(y_i | \boldsymbol{\theta})$$

of the log-likelihood contributions $l_i(\boldsymbol{\theta}) = \log f_i(y_i | \boldsymbol{\theta})$. This also implies that the score function and the Fisher information remain additive in terms of the observations as before.

Example B.3 Poisson Regression

We consider the following Poisson regression model:

$$\begin{aligned} Y_i &\sim \text{Po}(\lambda_i) \quad i = 1, \dots, n, \\ \lambda_i &= \exp(\beta_0 + \beta_1 x_i) = \exp(\eta_i), \\ \eta_i &= \beta_0 + \beta_1 x_i. \end{aligned}$$

Our aim is to compute the MLEs for β_0 and β_1 . It will turn out that an analytical solution in closed form as in Example B.1 is no longer available. Therefore, computation has to be based on numerical optimization methods, as seen in the following section. The other steps for determining the MLE are conceptually the same as before:

1. Likelihood

Omitting factors that do not depend on β_0 and β_1 , the likelihood contribution of observation i is

$$L_i(\beta_0, \beta_1) = \lambda_i^{y_i} \exp(-\lambda_i).$$

The likelihood of the sample is the product

$$L(\beta_0, \beta_1) = \prod_{i=1}^n L_i(\beta_0, \beta_1)$$

of individual likelihood contributions.

2. Log-likelihood

Taking logarithms, we obtain the individual log-likelihood contributions

$$l_i(\beta_0, \beta_1) = y_i \log(\lambda_i) - \lambda_i = y_i(\beta_0 + \beta_1 x_i) - \exp(\beta_0 + \beta_1 x_i)$$

and, summing up,

$$l(\beta_0, \beta_1) = \sum_{i=1}^n (y_i(\beta_0 + \beta_1 x_i) - \exp(\beta_0 + \beta_1 x_i)).$$

3. Score function

The partial first derivatives of individual log-likelihoods are

$$\begin{aligned} \frac{\partial l_i(\beta_0, \beta_1)}{\partial \beta_0} &= y_i - \exp(\beta_0 + \beta_1 x_i) = y_i - \lambda_i, \\ \frac{\partial l_i(\beta_0, \beta_1)}{\partial \beta_1} &= y_i x_i - \exp(\beta_0 + \beta_1 x_i) \cdot x_i = x_i(y_i - \lambda_i). \end{aligned}$$

Defining the vectors $\mathbf{y} = (y_1, \dots, y_n)'$, $\mathbf{x} = (x_1, \dots, x_n)'$, $\boldsymbol{\lambda} = (\lambda_1, \dots, \lambda_n)'$, $\mathbf{1} = (1, \dots, 1)'$ and the design matrix $\mathbf{X} = (\mathbf{1} \ \mathbf{x})$, we obtain the score function

$$s(\beta_0, \beta_1) = \begin{pmatrix} \sum_{i=1}^n (y_i - \lambda_i) \\ \sum_{i=1}^n x_i (y_i - \lambda_i) \end{pmatrix} = \begin{pmatrix} \mathbf{1}'(\mathbf{y} - \boldsymbol{\lambda}) \\ \mathbf{x}'(\mathbf{y} - \boldsymbol{\lambda}) \end{pmatrix} = \mathbf{X}'(\mathbf{y} - \boldsymbol{\lambda}).$$

Setting the score function to zero, we obtain a nonlinear system of equations that has to be solved numerically, as seen in the following section. The Fisher information matrix, determined in the next step, is an important component of the numerical algorithm.

4. *Fisher information matrix*

The second partial derivatives of the individual log-likelihoods are

$$\frac{\partial^2 l_i(\beta_0, \beta_1)}{\partial \beta_0^2} = -\exp(\beta_0 + \beta_1 x_i) = -\lambda_i,$$

$$\frac{\partial^2 l_i(\beta_0, \beta_1)}{\partial \beta_1^2} = -\exp(\beta_0 + \beta_1 x_i) x_i^2 = -\lambda_i x_i^2,$$

$$\frac{\partial^2 l_i(\beta_0, \beta_1)}{\partial \beta_0 \partial \beta_1} = -\exp(\beta_0 + \beta_1 x_i) x_i = -\lambda_i x_i.$$

Summing up and changing signs, we obtain the observed information matrix

$$\mathbf{H}(\boldsymbol{\beta}) = -\frac{\partial^2 l(\beta_0, \beta_1)}{\partial \boldsymbol{\beta} \partial \boldsymbol{\beta}'} = -\sum_{i=1}^n \frac{\partial^2 l_i(\beta_0, \beta_1)}{\partial \boldsymbol{\beta} \partial \boldsymbol{\beta}'} = \begin{pmatrix} \sum_{i=1}^n \lambda_i & \sum_{i=1}^n \lambda_i x_i \\ \sum_{i=1}^n \lambda_i x_i & \sum_{i=1}^n \lambda_i x_i^2 \end{pmatrix}.$$

Defining the diagonal matrix $\mathbf{W} = \text{diag}(\lambda_1, \dots, \lambda_n)$, we can write

$$\mathbf{H}(\boldsymbol{\beta}) = \mathbf{X}' \mathbf{W} \mathbf{X}$$

in matrix notation. As $\mathbf{H}(\boldsymbol{\beta})$ does not depend on $\mathbf{y}$, the observed information matrix is equal to the expected information matrix in this example, i.e.,

$$\mathbf{F}(\boldsymbol{\beta}) = \mathbf{E}_y(\mathbf{H}(\boldsymbol{\beta})) = \mathbf{H}(\boldsymbol{\beta}).$$

In general, however, $\mathbf{F}(\boldsymbol{\beta}) \neq \mathbf{H}(\boldsymbol{\beta})$.

5. *Iterative numerical computation of the MLE $\hat{\boldsymbol{\beta}}$*

$\hat{\boldsymbol{\beta}}$ is computed iteratively as the solution of the nonlinear system of equations

$$\mathbf{s}(\hat{\boldsymbol{\beta}}) \stackrel{!}{=} \mathbf{0}$$

using the Newton or Fisher scoring algorithm sketched in the following section. $\triangle$

We finally remark that the likelihood principle can also be applied if the variables $Y_1, \dots, Y_n$ are not independent. Such a situation appears, for example, in Sect. 7.3.2. There, the response vector $\mathbf{Y} = (Y_1, \dots, Y_n)'$ follows a multivariate normal distribution

$$\mathbf{Y} \sim \mathbf{N}(\boldsymbol{\mu}(\boldsymbol{\beta}), \boldsymbol{\Sigma}(\boldsymbol{\alpha})),$$

where the mean vector $\boldsymbol{\mu}(\boldsymbol{\beta})$ depends on $\boldsymbol{\beta}$ and the (non-diagonal) covariance matrix $\boldsymbol{\Sigma}(\boldsymbol{\alpha})$ depends on an unknown parameter vector $\boldsymbol{\alpha}$. The likelihood $L(\boldsymbol{\beta}, \boldsymbol{\alpha}) = L(\boldsymbol{\theta})$, $\boldsymbol{\theta} = (\boldsymbol{\alpha}, \boldsymbol{\beta})$ is then given by the p.d.f. of the multivariate normal distribution for $\mathbf{Y}$. It can no longer be factorized in the product of separate univariate p.d.f.'s for the variables $Y_1, \dots, Y_n$.

B.4.2 Numerical Computation of the MLE

In most applications, the MLE cannot be determined analytically: the system of equations, obtained from setting the score function to zero, is usually nonlinear and cannot be solved for the unknown parameters in closed form; see Example B.3. Therefore, numerical methods for computing the roots of the score function are required. Various such methods have been developed; see, for example, Lange (2000) for a survey. Here we only sketch the two most popular algorithms: the *Newton (or Newton–Raphson) method* and the *Fisher scoring algorithm*.

We first illustrate the Newton–Raphson method for a scalar parameter. The aim is to numerically compute the root of the score function, i.e., the solution of the generally nonlinear score equation

$$s(\theta) = 0.$$

Beginning with an initial value $\theta^{(0)}$, an (approximate) solution is computed iteratively as follows (see Fig. B.1): At $\theta^{(0)}$ the score function is approximated through a straight line, the *tangent*. An improved approximate solution $\theta^{(1)}$ is then obtained as the root of the tangent. The tangent is given by

$$g(\theta) = s(\theta^{(0)}) + s'(\theta^{(0)}) \cdot (\theta - \theta^{(0)}),$$

obtained through a first-order Taylor expansion of $s(\theta)$ at $\theta^{(0)}$. The root of the tangent provides the improved iterate

$$\theta^{(1)} = \theta^{(0)} - \frac{1}{s'(\theta^{(0)})} \cdot s(\theta^{(0)}).$$

Since $-s'(\theta)$ is the observed Fisher information $H(\theta)$, we can also write

$$\theta^{(1)} = \theta^{(0)} + H(\theta^{(0)})^{-1} s(\theta^{(0)}).$$

Starting from $\theta^{(1)}$ we get the next improved approximate solution $\theta^{(2)}$ by constructing another tangent of $s(\theta)$ at $\theta^{(1)}$ and computing its root $\theta^{(2)}$. This algorithm continues iteratively until successive iterates do practically not change any more.

The extension of the algorithm to a multivariate parameter $\theta = (\theta_1, \dots, \theta_p)'$ is as follows: Let $\theta^{(t)}$ be the current iterate of the approximate solution of $s(\theta) = \mathbf{0}$. An improved iterate is

$$\theta^{(t+1)} = \theta^{(t)} - \left(\frac{\partial s(\theta^{(t)})}{\partial \theta'} \right)^{-1} s(\theta^{(t)}) = \theta^{(t)} + H(\theta^{(t)})^{-1} s(\theta^{(t)}).$$

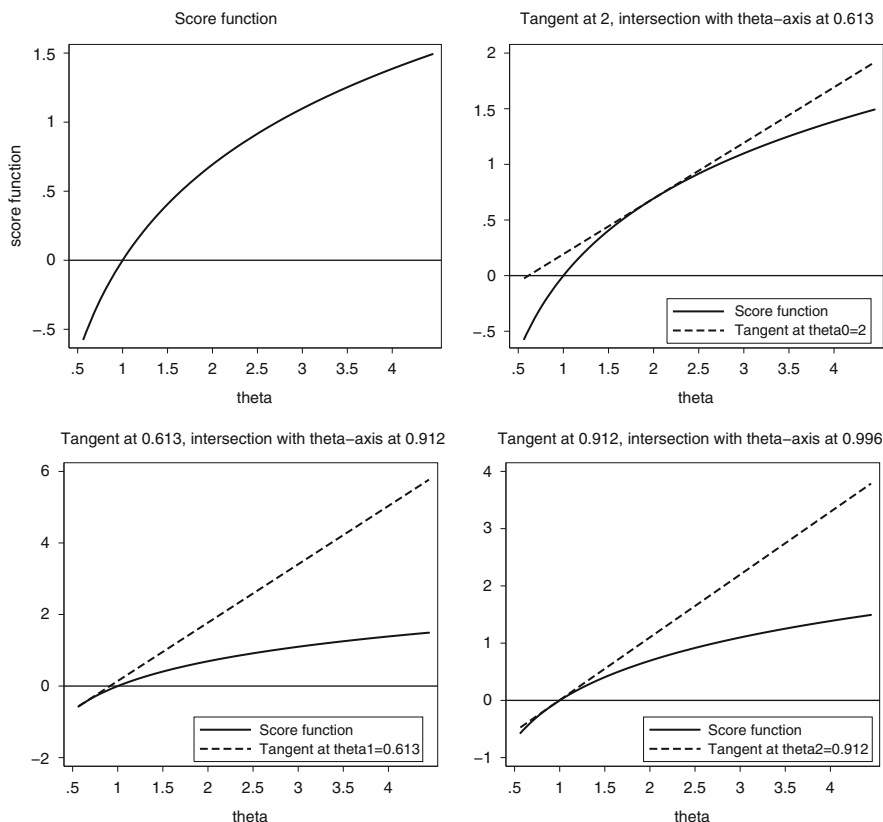


Fig. B.1 Illustration of the Newton method

The *Fisher scoring method* is obtained if the observed information matrix is replaced by the expected information matrix $F(\theta^{(t)})$. As an advantage, the expression for the expected information matrix is often simpler and faster to compute.

Example B.4 Poisson Regression

Continuing the Poisson regression example, the Newton–Raphson method for computing the MLE $\hat{\beta} = (\hat{\beta}_0, \hat{\beta}_1)'$ works as follows:

1. Define the initial values $\hat{\beta}^{(0)}$, e.g., $\hat{\beta}^{(0)} = (0, 0)'$. Set $t = 0$.
2. Compute improved estimates $\hat{\beta}^{(t)}$ through

$$\hat{\beta}^{(t+1)} = \hat{\beta}^{(t)} + H(\hat{\beta}^{(t)})^{-1} s(\hat{\beta}^{(t)}) = \hat{\beta}^{(t)} + (X'WX)^{-1} X'(y - \lambda),$$

where $W = W(\hat{\beta}^{(t)})$ and $\lambda = \lambda(\hat{\beta}^{(t)})$ depend on the current iterate $\hat{\beta}^{(t)}$.

3. Stop if

$$\frac{\|\hat{\beta}^{(t+1)} - \hat{\beta}^{(t)}\|}{\|\hat{\beta}^{(t)}\|} \leq \varepsilon$$

for a (very) small value $\varepsilon > 0$. Otherwise set $t = t + 1$ and go to step 2.

△

B.4.3 Asymptotic Properties of the MLE

Under certain regularity assumptions, depending in detail on the specific model $f(y_1, \dots, y_n | \theta)$, the following asymptotic properties hold: For $n \rightarrow \infty$, or practically for large sample size, the MLE is consistent, asymptotically (or approximately) unbiased, and asymptotically (or approximately) multivariate normal:

$$\hat{\theta} \stackrel{a}{\sim} N(\theta, \hat{V}).$$

The (estimated) covariance matrix $\widehat{\text{Cov}}(\hat{\theta}) = \hat{V}$ is obtained as the inverse of the observed or expected information matrix, i.e.,

$$\hat{V} = H^{-1}(\hat{\theta}) \quad \text{or} \quad \hat{V} = F^{-1}(\hat{\theta}),$$

evaluated at the MLE $\theta = \hat{\theta}$. These inverses are computed in the final iteration step of the Newton or Fisher scoring algorithm. The diagonal elements $\hat{v}_{jj}$ of $\hat{V}$ are the estimated variances $\widehat{\text{Var}}(\hat{\theta}_j)$ of the j th component of $\hat{\theta}$, and $\sqrt{\hat{v}_{jj}}$ is the estimated standard deviation or standard error: $se_j = \sqrt{\hat{v}_{jj}}$.

Finally, MLEs are also asymptotically efficient, i.e., for large sample sizes their variance is smaller or at least not larger than the variance of other asymptotically unbiased estimators.

How large should the sample size n be so that these asymptotic properties hold sufficiently well? It is not possible to give a general definitive answer to this question. Basically, it depends on the quality of the approximation of the likelihood or the log-likelihood in a neighborhood of $\hat{\theta}$ through a normal distribution or a quadratic function, respectively. As a rule of thumb, the sample size n has to be much larger (about a factor of 10–20) than the number p of unknown parameters. If in doubt, simulation studies are useful.

B.4.4 Likelihood-Based Tests of Linear Hypotheses

We consider testing linear hypotheses about $\theta = (\theta_1, \dots, \theta_p)' \in \Theta$ of the form

$$H_0 : C\theta = d \quad \text{versus} \quad H_1 : C\theta \neq d,$$

where the $(r \times p)$ -matrix C has full row rank $r \leq p$. In this book θ is mostly the vector β of regression coefficients. An important special case are hypotheses about an r -dimensional subvector θ_1 of θ of the form

$$H_0 : \theta_1 = d \quad \text{versus} \quad H_1 : \theta_1 \neq d.$$

If $\theta_1 = \beta_1$ is a subvector of regression coefficients and $\mathbf{d} = \mathbf{0}$, this is a test about significance of corresponding covariates. The three commonly used test statistics are the (log-)likelihood ratio statistic, the Wald statistic, and the score statistic.

The *(log-)likelihood ratio statistic*

$$lr = 2\{l(\hat{\theta}) - l(\tilde{\theta})\} = -2\{l(\tilde{\theta}) - l(\hat{\theta})\}$$

compares the unrestricted maximum $l(\hat{\theta})$ of the log-likelihood with the maximum obtained for the restricted MLE $l(\tilde{\theta})$, computed under the restriction $\mathbf{C}\theta = \mathbf{d}$ of H_0 . Since $l(\hat{\theta}) \geq l(\tilde{\theta})$, we have $lr \geq 0$. If the unrestricted maximum $l(\hat{\theta})$ is significantly larger than $l(\tilde{\theta})$, implying that lr is large, H_0 will be rejected in favor of H_1 . The test statistic can also be motivated by considering the likelihood ratio

$$LR = \frac{L(\hat{\theta})}{L(\tilde{\theta})},$$

where $L(\hat{\theta})$ and $L(\tilde{\theta})$ are the maximum of the likelihood without and under H_0 , respectively. Now $LR \geq 1$, and H_0 will be rejected if LR is significantly larger than 1. Taking logarithms and multiplying by the factor 2, one obtains the (log-)likelihood ratio test statistic.

To compute lr , the log-likelihood $l(\theta)$ has to be maximized under the restriction of H_0 , which can require considerable effort for the general form $\mathbf{C}\theta = \mathbf{d}$ of the null restriction. This can be avoided with the *Wald statistic*

$$w = (\mathbf{C}\hat{\theta} - \mathbf{d})'(\mathbf{C}\hat{\mathbf{V}}\mathbf{C}')^{-1}(\mathbf{C}\hat{\theta} - \mathbf{d}).$$

It measures the weighted distance between the unrestricted estimate $\mathbf{C}\hat{\theta}$ of $\mathbf{C}\theta$ and its hypothetical value $\mathbf{d}$ under H_0 . The weight is the inverse of the (estimated) covariance matrix $\mathbf{C}\hat{\mathbf{V}}\mathbf{C}'$ of $\mathbf{C}\hat{\theta} - \mathbf{d}$, where $\hat{\mathbf{V}}$ is the (estimated) covariance matrix of $\hat{\theta}$. If H_0 is true, this weighted distance should be small, whereas H_0 should be rejected if w is large. For the special hypothesis $H_0 : \theta_1 = \mathbf{d}$, the Wald statistic simplifies to

$$w = (\hat{\theta}_1 - \mathbf{d})'\hat{\mathbf{V}}_1^{-1}(\hat{\theta}_1 - \mathbf{d}),$$

where $\hat{\mathbf{V}}_1 = \widehat{\text{Cov}}(\hat{\theta}_1)$ is the (estimated) covariance matrix of $\hat{\theta}_1$.

The *score statistic*

$$u = s'(\tilde{\theta})\tilde{\mathbf{V}}s(\tilde{\theta})$$

measures the weighted distance between the value $\mathbf{0} = s(\hat{\theta})$ of the score function at the MLE $\hat{\theta}$ and the value $s(\tilde{\theta})$ at the restricted MLE $\tilde{\theta}$, where $\tilde{\mathbf{V}}$ is the estimated covariance matrix of $\tilde{\theta}$.

Wald tests are computationally advantageous in cases when H_0 defines a submodel that has to be tested against a larger model (without the H_0 restriction)

that has been estimated already. Additionally, score tests are advantageous if an already fitted model has to be tested against a larger model. Therefore, numerically efficient forward and backward variable selection may be based on Wald and score tests.

Under H_0 , the three test statistics are asymptotically equivalent and asymptotically (or approximately) chi-squared distributed with r degrees of freedom:

$$lr, w, s \stackrel{a}{\sim} \chi_r^2.$$

Critical values or p-values are computed using the limiting χ_r^2 -distribution. For medium and large sample sizes, this approximation is usually sufficient. For smaller sample sizes, the three test statistics may be comparably different from each other. Note that the Wald test is not restricted to genuine likelihood inference. It is applicable whenever an estimator $\hat{\theta}$ is asymptotically normal, as, for example, in least squares inference without normality assumption or in quasi-likelihood inference.

B.4.5 Model Choice

For comparing and choosing between several competing statistical models with different predictors and parameters, we need a compromise between good fit to the data and model complexity, i.e., a (too) high effective number of parameters. In linear regression models, for example, the goodness-of-fit measure R^2 will be increased by including additional covariates, interactions, etc. However, this usually results in overfitting the (training) data and worsens predictive capabilities and generalization to new data. Therefore, criteria for goodness of fit, such as R^2 and, for likelihood inference, the deviance or the log-likelihood, have to be modified such that overfitting the data is avoided by penalizing high model complexity, i.e., a high effective number of parameters. For likelihood inference with p -dimensional parameter $\theta = (\theta_1, \dots, \theta_p)'$, Akaike's information criterion AIC is defined as

$$\text{AIC} = -2l(\hat{\theta}) + 2p,$$

where the term $2p$ penalizes overly complex models. For model choice, models with small AIC value (and not with large log-likelihood $l(\hat{\theta})$!) are favored. Instead of the AIC, also AIC/n , where n is the sample size, is used.

For non- and semiparametric models as in Chaps. 8 and 9, the fixed dimension p is replaced by the “effective” dimension

$$df = \text{tr}(\mathbf{S}),$$

where $\mathbf{S}$ is the smoother matrix (see Sect. 8.1.8, p. 473). Alternative penalty terms, for example, the adjusted AIC_{corr} ,

$$\text{AIC}_{\text{corr}} = -2l(\hat{\theta}) + \frac{2n(df + 1)}{n - df - 2},$$

are also suggested. [Burnham and Anderson \(2002\)](#) give a nice justification of the AIC.

B.5 Bayesian Inference

Due to the development of computer intensive and simulation-based Markov chain Monte Carlo (MCMC) methods, Bayesian inference has become an attractive approach for analyzing complex statistical models. We first introduce concepts of Bayesian inference and then describe MCMC methods. More detailed introductions can be found, e.g., in [Migon and Gamerman \(1999\)](#) and [Held and Sabanés Bové \(2012\)](#).

B.5.1 Basic Concepts of Bayesian Inference

The fundamental difference to likelihood-based inference is that the unknown parameters $\theta = (\theta_1, \dots, \theta_p)'$ are not considered as fixed, deterministic quantities but as random variables with a *prior distribution*. A Bayesian model therefore consists of two parts:

- *Prior distribution*: Any (subjective) information about the unknown parameter θ is expressed by specifying a probability distribution for θ . This distribution is called prior (distribution) of θ . The specification of a prior for θ does not necessarily mean that the unknown parameters are actually stochastic quantities. Rather the prior describes the *degree of uncertainty* about the unknown parameters prior to the statistical analysis. The p.d.f. or probability function of the prior distribution will be denoted by $p(\theta)$.
- *Observation model*: The observation model specifies the conditional distribution of observable quantities, that is, the random sample variables $\mathbf{Y} = (Y_1, \dots, Y_n)'$, given the parameters. The p.d.f. or probability function of this conditional distribution is proportional to the likelihood $L(\theta)$ and will be denoted by $p(\mathbf{y} | \theta)$.

Based on the prior and the observation model, Bayes' theorem determines the distribution of θ after the data are known through the statistical experiment, that is the conditional distribution of θ given the observations $\mathbf{y} = (y_1, \dots, y_n)'$. We obtain

$$p(\theta | \mathbf{y}) = \frac{p(\mathbf{y} | \theta) p(\theta)}{\int p(\mathbf{y} | \theta) p(\theta) d\theta} = c \cdot p(\mathbf{y} | \theta) p(\theta),$$

with the normalizing constant $c = [\int p(\mathbf{y} | \theta) p(\theta) d\theta]^{-1}$. This conditional distribution is called *posterior (distribution)*.

Example B.5 Poisson Distribution

Consider an i.i.d. sample $Y_1, \dots, Y_n$ from a Poisson distribution, i.e., $Y_i \sim \text{Po}(\lambda)$. The unknown parameter is to be estimated using Bayesian inference. As in Example B.1, the joint probability for the observed sample $\mathbf{y} = (y_1, \dots, y_n)'$ is

$$p(\mathbf{y} | \lambda) = \frac{1}{y_1! \cdots y_n!} \lambda^{\sum_{i=1}^n y_i} \exp(-n\lambda).$$

We specify a gamma distribution with parameters a and b for λ , i.e., $\lambda \sim G(a, b)$. A justification for the choice of this prior will be given later. It follows that λ has p.d.f.

$$p(\lambda) = k \lambda^{a-1} \exp(-b\lambda)$$

with $k = \frac{b^a}{\Gamma(a)}$. The posterior is obtained as

$$p(\lambda | \mathbf{y}) = \frac{p(\mathbf{y} | \lambda) p(\lambda)}{\int p(\mathbf{y} | \lambda) p(\lambda) d\lambda} = c \frac{1}{y_1! \cdots y_n!} \lambda^{\sum_{i=1}^n y_i} \exp(-n\lambda) k \lambda^{a-1} \exp(-b\lambda).$$

To determine the type of this distribution, we can ignore all factors that do not depend on λ . This gives

$$p(\lambda | \mathbf{y}) \propto \lambda^{\sum_{i=1}^n y_i} \exp(-n\lambda) \lambda^{a-1} \exp(-b\lambda) = \lambda^{a + \sum_{i=1}^n y_i - 1} \exp(-(b + n)\lambda).$$

This has the form of a gamma distribution with parameters $a' = a + \sum_{i=1}^n y_i$ and $b' = b + n$, i.e.,

$$\lambda | \mathbf{y} \sim G\left(a + \sum_{i=1}^n y_i, b + n\right),$$

and the posterior has the same type of distribution as the prior. This desirable case appears in many other simple Bayesian models. We call the prior as conjugate to the Poisson model because the posterior is of the same type as the prior. $\triangle$

The next example discusses a situation where we have rather limited prior knowledge about the unknown parameters. In this case we are confronted with the question how to specify a prior in the absence of prior knowledge. We call such priors *noninformative*.

Example B.6 Bayesian Logit Model—Diffuse Prior

We consider a logit model with a single covariate x :

$$Y_i = B(1, \pi_i), \quad \pi_i = \frac{\exp(\eta_i)}{1 + \exp(\eta_i)}, \quad \eta_i = \beta_0 + \beta_1 x_i, \quad i = 1, \dots, n.$$

Assuming, as usual, (conditionally) independent response variables, the observation model is given by

$$p(\mathbf{y} | \boldsymbol{\beta}) \propto L(\boldsymbol{\beta}) = \prod_{i=1}^n \pi_i^{y_i} (1 - \pi_i)^{1-y_i},$$

where $\boldsymbol{\beta} = (\beta_0, \beta_1)'$ is the vector of regression coefficients. To fully specify the Bayesian model, we have to supplement the observation model by a prior for β_0 and β_1 .

Since estimated regression coefficients are often approximately normally distributed, it is reasonable to assume a two-dimensional normal prior. Therefore, we specify the prior

$$p(\boldsymbol{\beta}) \sim N(\mathbf{m}, \mathbf{M})$$

with prior mean $\mathbf{m}$ and prior covariance matrix $\mathbf{M}$. If, for example, results from a previous statistical analysis are available, we could choose the previous point estimate as $\mathbf{m}$ and its estimated covariance matrix as $\mathbf{M}$. If the previous analysis has been carried out some time ago, we may also multiply $\mathbf{M}$ with a factor $a > 1$ to express increased uncertainty.

In many situations, however, we may have no information about the regression coefficients. Increasing the variances in $\mathbf{M}$, the normal prior becomes very flat and approximates a uniform distribution. In the limiting case the prior becomes proportional to a constant, i.e.,

$$p(\boldsymbol{\beta}) \propto \text{const.}$$

Often we also write $p(\boldsymbol{\beta}) \propto 1$. The integral of this flat prior over $\mathbb{R}^2$ is not finite, so that $p(\boldsymbol{\beta})$ is not a density in the usual sense. Such a prior is called improper or diffuse. Nonetheless such diffuse priors are admissible as long as the posterior, resulting from Bayes' theorem, is a proper distribution, i.e., its integral over $\mathbb{R}^2$ is finite. In a Bayesian logit model this is the case if a finite MLE exists. With a flat, diffuse prior the posterior density is

$$p(\boldsymbol{\beta} | y) = \frac{p(\boldsymbol{\beta})p(y | \boldsymbol{\beta})}{\int p(\boldsymbol{\beta})p(y | \boldsymbol{\beta})d\boldsymbol{\beta}} \propto p(y | \boldsymbol{\beta}) = \prod_{i=1}^n \pi_i^{y_i} (1 - \pi_i)^{1-y_i}.$$

Although the posterior is proper, it has no known distributional type. This causes problems for statistical inference because characteristics of the posterior, such as the mean, are not available in closed analytical form. However inference based on MCMC simulation is possible. See Sect. B.5.3 and in particular, Example B.10. $\triangle$

The previous example suggests to assume a uniform prior distribution for the parameters if there is no prior knowledge. The next example, however, shows that the construction of noninformative priors is a bit more subtle.

Example B.7 Normal Distribution

Consider an i.i.d. sample $Y_1, \dots, Y_n$ from a normal distribution, i.e., $Y_i \sim N(\mu, \sigma^2)$. We first assume the expectation μ to be known. In the absence of any prior knowledge regarding the unknown parameter σ^2 one is tempted to assume a diffuse uniform prior over $\mathbb{R}^+$. It follows that $p(\sigma^2)$ is improper and proportional to a constant, i.e., $p(\sigma^2) \propto 1$. Suppose now, that we had parameterized the normal distribution in terms of the standard deviation σ rather than the variance. According to our recipe for a noninformative prior we would then assign a uniform distribution for σ , i.e., $p(\sigma) \propto 1$. From this assumption, we can derive the corresponding distribution of the variance σ^2 using the change of variables theorem of continuous random variables (Theorem B.1 in Appendix B.1). We now obtain $p(\sigma^2) \propto (\sigma^2)^{-1/2}$, which is no longer a uniform distribution over $\mathbb{R}^+$. Thus, if we parameterize the normal distribution with the variance our noninformative prior for σ^2 is a uniform distribution. If we parameterize in terms of the standard deviation we come out with a nonuniform distribution for σ^2 . This means that the prior is not invariant with respect to the specific parameterization. $\triangle$

A class of priors that help to circumvent the invariance problem mentioned in the example are *Jeffreys' priors*. Jeffreys' prior for a scalar parameter θ is defined to be proportional to the square root of the expected Fisher information, i.e.,

$$p(\theta) \propto \sqrt{F(\theta)} = \sqrt{E(-l''(\theta))}.$$

Jeffreys' prior solves the invariance problem as it is indeed invariant with respect to one-to-one transformations of the parameter. The remaining question is if Jeffreys' prior is also noninformative in some sense. The answer is yes, as Jeffreys' prior for a scalar parameter can be characterized as a *reference prior*. The concept of a reference prior has been introduced in [Bernardo \(1979\)](#). Informally, the reference prior can be characterized as the distribution that maximizes the influence of the data on the posterior. More specifically, the reference prior maximizes the expected Kullback–Leibler distance of the posterior relative to the prior. In this sense the reference prior is noninformative as the data get maximal weight and the influence of the prior is minimized.

In situations with vector-valued parameter θ , Jeffreys' prior generalizes to

$$p(\theta) \propto \sqrt{|F(\theta)|}.$$

This prior is still translation invariant, but is in general not the reference prior. This is one of the reasons why Jeffreys' prior is usually not accepted as a valid noninformative prior for vector valued parameters. See, e.g., [Held and Sabanés Bové \(2012\)](#) for a discussion of the deficiencies of Jeffreys' prior in the multiparameter case. Instead, we can still base the choice of noninformative priors on the reference prior concept. However, it would be beyond the scope of this book to fully introduce this technically difficult concept; see, e.g., [Held and Sabanés Bové \(2012\)](#) for details.

Example B.8 Normal Distribution—Jeffreys' Prior—Reference Prior

We continue the previous Example [B.7](#) and derive the Jeffreys' or reference prior. In the case of a normal distribution with known expectation μ , Jeffreys' prior or the reference prior for σ^2 is easily obtained as

$$p(\sigma^2) \propto \frac{1}{\sigma^2}.$$

Now the distribution of the standard deviation can be derived to be $p(\sigma) \propto 1/\sigma$ which is the same distribution as for σ^2 .

If both parameters μ and σ^2 are unknown we obtain a noninformative prior in the form of the reference prior. This reference prior is given by

$$p(\mu, \sigma^2) \propto \frac{1}{\sigma^2}$$

implying a priori independence of μ and σ^2 with distributions $p(\mu) \propto 1$ and $p(\sigma^2) \propto 1/\sigma^2$.

△

B.5.2 Point and Interval Estimation

Point Estimation

The usual point estimators in Bayesian inference are the posterior mean, the posterior median, and the posterior mode. These three estimators can be discussed and justified from a decision theoretic perspective. We do not describe this here in detail; rather we only introduce the estimators.

The posterior mean is given by

$$\hat{\theta} = E(\theta | y) = \int \theta p(\theta | y) d\theta = c \cdot \int \theta p(y | \theta) p(\theta) d\theta.$$

Thus, the analytical or numerical determination of the posterior mean requires the computation of (possibly high-dimensional) integrals. The resulting difficulties have been a main obstacle for applying Bayesian methods in practice. Simulation-based methods, in particular modern MCMC methods, have greatly reduced these problems. Refer to Sect. B.5.3.

The posterior mode is the value $\hat{\theta}$ that (globally) maximizes the posterior density, i.e.,

$$\hat{\theta} = \arg \max_{\theta} p(\theta | y) = \arg \max_{\theta} p(y | \theta) p(\theta).$$

The second expression shows that no integration is necessary to compute the posterior mode, because the normalizing constant is not needed.

The posterior median, that is, the median of the posterior distribution, is sometimes preferred to the posterior mean because it is more robust against outliers.

Example B.9 Poisson Distribution

In Example B.5 we obtained a gamma distribution with parameters $a' = a + \sum_{i=1}^n y_i$ and $b' = b + n$ as the posterior for λ . The posterior mean is

$$E(\lambda | y) = \frac{a + \sum_{i=1}^n y_i}{b + n}.$$

The smaller a (in relation to $\sum y_i$) and b (in relation to n), the closer the posterior mean is to the usual MLE $\hat{\lambda} = \bar{y}$. The larger the prior information, that is, the larger a and b are, the more the posterior mean and the MLE differ from each other.

△

Interval Estimation

Point estimators reduce the information of the posterior to a single number. In particular, it is not possible to see how precise these estimates are. Usual quantities for measuring variability of a random variable are also natural measures for assessing how precise, or imprecise, the estimators are. For the posterior mean, a natural measure is the posterior variance; for the posterior median, the interquartile distance seems to be appropriate to measure its variability. In case of the posterior mode, the

curvature of the posterior at the mode, i.e., the observed Fisher information, is a natural choice.

Another popular way of assessing uncertainty is Bayesian confidence intervals or *credible intervals* or, more generally, *credible regions*:

A region $C \subset \Theta$ of the parameter space is said to be a $(1 - \alpha)$ -credible region for θ if

$$P(\theta \in C \mid \mathbf{y}) = 1 - \alpha.$$

If $C \subseteq \mathbb{R}$ is an interval it is called credible interval. In other words, a credible region contains (at least) a probability mass $1 - \alpha$ of the posterior. Note that credible regions and classical confidence regions have rather different interpretations: The classical confidence interval does not provide a probability statement about θ , which is a deterministic quantity in the classical inferential concept. Rather, it is related with a probability statement about the random sample $\mathbf{y} = (y_1, \dots, y_n)'$: The confidence region $C(\mathbf{y})$ contains the unknown true parameter θ with probability $1 - \alpha$. The frequentist interpretation is that confidence intervals cover the true parameter with a percentage of about $(1 - \alpha) \cdot 100\%$ if the estimation procedure is repeated for many samples. In contrast, Bayesian credible intervals allow a probability statement about the random parameter θ : The credible region contains $(1 - \alpha) \cdot 100\%$ of the probability mass of the posterior, i.e., a random number θ drawn from the posterior is within the credible region with probability $1 - \alpha$.

Bayesian credible intervals are often difficult to determine analytically. However, they can be easily computed with MCMC algorithms for drawing random numbers from the posterior.

B.5.3 MCMC Methods

Historically, the main difficulty in applying Bayesian methods has been that in many situations the posterior is not available analytically or even numerically. Basically, MCMC is Monte Carlo integration using Markov chains. Although MCMC was known in physics very early ([Metropolis, Rosenbluth, Rosenbluth, Teller, & Teller, 1953](#)), it took nearly 40 years to enter mainstream statistical practice ([Gelfand & Smith, 1990](#)), in particular for Bayesian inference. Since then, MCMC has had an enormous effect on Bayesian statistics, and it allows to solve complex and high-dimensional inferential problems. Good introductions are, for example, [Gilks, Richardson, and Spiegelhalter \(1996\)](#) and [Green \(2001\)](#). Here we only describe the basic idea and the most important algorithms.

MCMC methods allow to draw samples from posterior distributions (and, in principle, from any distribution) that are usually not available analytically and to estimate characteristics of the posterior such as the mean, the variance or quantiles, or the posterior density itself. The most important advantage compared to more traditional methods of drawing a sample from a distribution, for example, importance or rejection sampling, is that samples can be drawn from high-dimensional densities, even for dimensions in the thousands. Another advantage is that the normalizing

constant, often a high-dimensional integral that cannot be computed with traditional numerical methods, does not have to be known.

The basic idea of MCMC methods is comparably simple. Let θ be the unknown vector of parameters in a Bayesian model and $p(\theta | y)$ the posterior density (we assume here that θ is continuous). Instead of directly drawing an i.i.d. sample from $p(\theta | y)$, a Markov chain is generated such that the iterations of the transition kernel converge to the posterior of interest. In this way random numbers are generated that can be considered as a (correlated) sample from the posterior after some time of convergence, the *burn-in phase*. Before describing some algorithms in detail, we point out that MCMC may be used for drawing random numbers from any complex distribution, not only from posterior distributions. The posterior $p(\theta | y)$ then has to be replaced by such a density in the following.

Metropolis–Hastings Algorithm

The basic algorithm, from which all other algorithms can be derived, works as follows: First, a starting value $\theta^{(0)}$ is chosen. Instead of drawing directly from the posterior $p(\theta | y)$, a new random number θ^* is drawn from a *proposal density* q in each iteration. Usually, the proposal density depends on the current iterate (or state) $\theta^{(t-1)}$, i.e., $q = q(\theta^* | \theta^{(t-1)})$. In principle, the choice of the proposal density is arbitrary. However, it should be comparably easy to draw random numbers from this density. Since the proposal density differs from the posterior, a proposed random number θ^* cannot always be accepted as the new current state $\theta^{(t)}$ but only with some acceptance probability α . Basically, this is the ratio of the posterior and the proposal density, evaluated at the current state $\theta^{(t-1)}$ and the proposed value θ^* . More specifically, the acceptance probability is defined as

$$\alpha(\theta^* | \theta^{(t-1)}) = \min \left\{ \frac{p(\theta^* | y) q(\theta^{(t-1)} | \theta^*)}{p(\theta^{(t-1)} | y) q(\theta^* | \theta^{(t-1)})}, 1 \right\}.$$

If the proposed θ^* is accepted, then the next state is $\theta^{(t)} = \theta^*$; otherwise $\theta^{(t)} = \theta^{(t-1)}$. The density $p(\theta | y)$ does appear directly, not only in $\alpha(\theta^* | \theta^{(t-1)})$, but also in the ratio $p(\theta^* | y)/p(\theta^{(t-1)} | y)$. Therefore, all expressions in $p(\theta | y)$ that are constant can be omitted. In particular, the normalizing constant of the posterior is not needed. This is one of the big advantages of MCMC methods compared to conventional methods for drawing random numbers. A summary of the MH algorithm is given in Box 2.1.

The Metropolis–Hastings algorithm simplifies for symmetric proposals with $q(\theta^* | \theta^{(t-1)}) = q(\theta^{(t-1)} | \theta^*)$. The acceptance probability then becomes

$$\alpha(\theta^* | \theta^{(t-1)}) = \min \left\{ \frac{p(\theta^* | y)}{p(\theta^{(t-1)} | y)}, 1 \right\}.$$

This is the original Metropolis algorithm, published in 1953, and generalized to nonsymmetric proposal densities by [Hastings \(1970\)](#).

2.1 Metropolis–Hastings Algorithm

To draw random numbers from the density $p(\boldsymbol{\theta} \mid \mathbf{y})$, the Metropolis–Hastings algorithm proceeds as follows:

1. Initialize $\boldsymbol{\theta}^{(0)}$ and the number T of iterations. Set $t = 1$.
2. Draw a random number $\boldsymbol{\theta}^*$ from the proposal density $q(\boldsymbol{\theta}^* \mid \boldsymbol{\theta}^{(t-1)})$ and accept it as the new state $\boldsymbol{\theta}^{(t)}$ with probability $\alpha(\boldsymbol{\theta}^* \mid \boldsymbol{\theta}^{(t-1)})$; otherwise set $\boldsymbol{\theta}^{(t)} = \boldsymbol{\theta}^{(t-1)}$.
3. Stop if $t = T$; otherwise set $t = t + 1$ and go to 2.

After a *burn-in phase* t_0 , the random numbers $\boldsymbol{\theta}^{(t_0+1)}, \dots, \boldsymbol{\theta}^{(T)}$ can be considered as a (correlated) sample from $p(\boldsymbol{\theta} \mid \mathbf{y})$.

Note that numerical computation of α is usually problematic in practice because the densities contain exponential functions. It is therefore more favorable to base acceptance/rejection on $\log(\alpha)$. Then in practice we generate a random number $u \sim U(0, 1)$ (uniform on $(0, 1)$) and accept $\boldsymbol{\theta}^*$ if $\log(u) \leq \log(\alpha)$ thereby avoiding the evaluation of exponential functions.

Although any proposal density will ultimately deliver samples from the posterior, it is important to choose appropriate proposal densities. In particular, acceptance probabilities should be large enough and correlation between successive iterates should be small. The smaller the correlation, the smaller is the required sample size to estimate characteristics of the posterior. We illustrate the construction of a Metropolis–Hastings algorithm for a Bayesian logit model.

Example B.10 Bayesian Logit Model

We consider the following simulated logit model with two covariates x_1 and x_2 :

$$Y_i = B(1, \pi_i) \quad i = 1, \dots, 500,$$

$$\pi_i = \frac{\exp(\eta_i)}{1 + \exp(\eta_i)},$$

$$\eta_i = -0.5 + 0.6 x_{i1} - 0.3 x_{i2}.$$

The covariates x_1 and x_2 are drawn independently from a standard normal distribution. We want to construct a Metropolis–Hastings algorithm to estimate the parameter $\boldsymbol{\beta} = (-0.5, 0.6, -0.3)'$ given this simulated data.

We specify independent diffuse priors $p(\beta_j) \propto \text{const}$. The posterior is then proportional to the likelihood:

$$p(\boldsymbol{\beta} \mid \mathbf{y}) \propto \prod_{i=1}^{500} \pi_i^{y_i} (1 - \pi_i)^{1-y_i}.$$

As a proposal density for the Metropolis–Hastings algorithm we choose a three-dimensional normal distribution, with the current state $\boldsymbol{\beta}^{(t-1)}$ as its mean. For its covariance matrix,

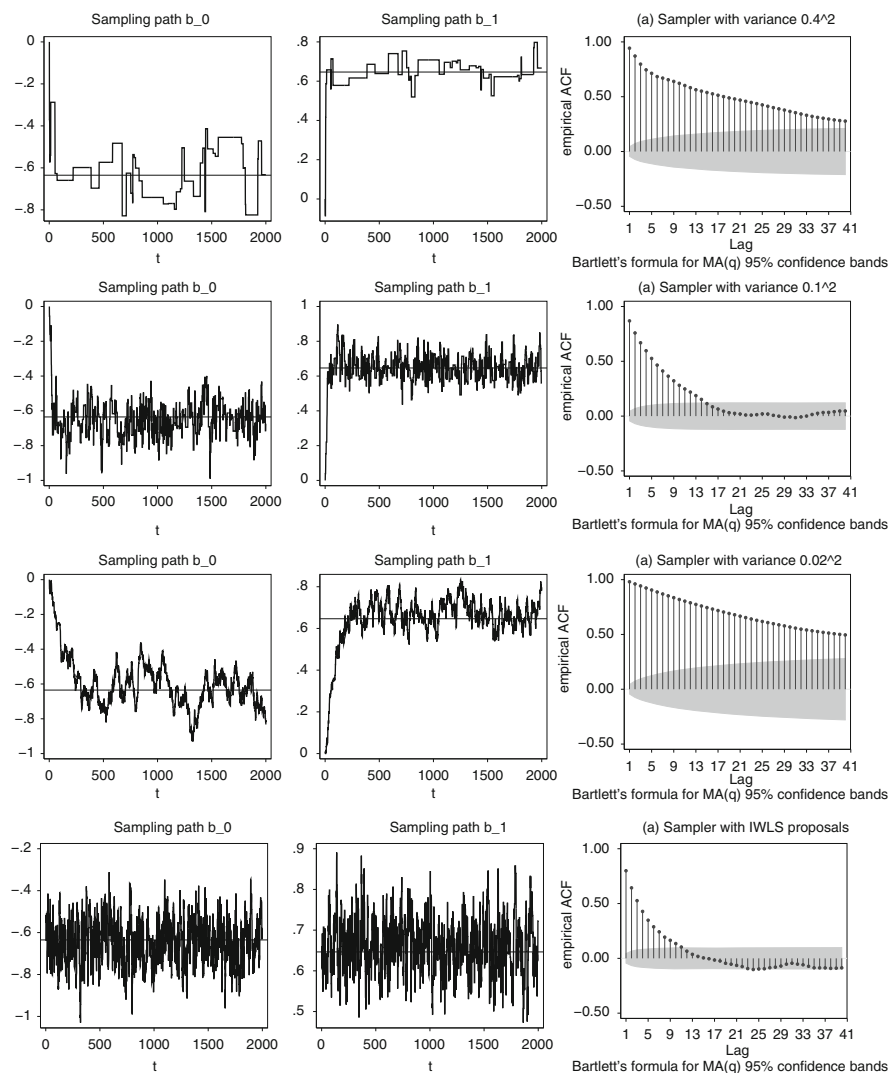


Fig. B.2 Sampling paths for β_0 and β_1 for different MH algorithms. The *third column* shows the respective autocorrelation functions for β_1

we start with the diagonal matrix $\Sigma = \text{diag}(0.4^2, 0.4^2, 0.4^2)$. Figure B.2 (first row) shows the first 2,000 random numbers for β_0 and β_1 drawn with this proposal density. Since we have specified diffuse priors, Bayes estimates for the regression coefficients should not differ too much from the MLEs. Therefore, the MLEs are displayed as horizontal lines in the plots. Clearly, only a few of the proposed random numbers are accepted with this first algorithm; sometimes the state remains unchanged for more than 100 iterations. Thus, the acceptance probabilities are far too small. We obtain larger acceptance probabilities if the variances of the proposal density are decreased to $\Sigma = \text{diag}(0.1^2, 0.1^2, 0.1^2)$. The second

row in Fig. B.2 shows the first 2,000 random numbers for β_0 and β_1 resulting from this second MH algorithm. We recognize a short burn-in phase of about 50 iterations, followed by reasonable iterations with relatively large acceptance rates. If we further decrease the variance to $\Sigma = \text{diag}(0.02^2, 0.02^2, 0.02^2)$, acceptance rates are further increased, but successive draws remain almost in the same state; see the third row in Fig. B.2.

A useful and important tool for assessing the quality of MCMC algorithms is the autocorrelation function of the sample [see section “First Order Autocorrelation” of Sect. 4.1.4 (p. 192), for the autocorrelation function]. Ideally, autocorrelations should rapidly converge to zero with increasing lags. The smaller the autocorrelation of successive parameters, the better the characteristics of the posterior can be estimated, based on the same length T of the sample. As an example, the right column of Fig. B.2 shows the autocorrelation functions for β_1 , corresponding to the three MH algorithms. Obviously, correlations for the first and third algorithm are extremely large, even for a lag of 40. On the other hand, autocorrelation for lags larger than 20 are almost zero for the second algorithm. For practical work, “thinning” is carried out for the original sample, i.e., only every k th random number is kept in the sample, so that the remaining random numbers are almost uncorrelated. In this way, memory space can be saved without worsening estimation results. To generate an approximately uncorrelated sample of size 1,000 with our second MH algorithm, we would have to generate about 20,000 random numbers after a short burn-in phase and then keep only every 20th random number in the thinned sample.

We can conclude the following: Small variances of the proposal density lead to high acceptance rates. In contrast, acceptance rates become small for large variances. For very large or very small variances autocorrelations of successive random numbers are high. The art of designing good MH algorithms is therefore the choice of appropriate proposal densities that combine high acceptance rates with low autocorrelations. Furthermore, automated methods are desirable that do not require subjective tuning of parameters of the proposal density.

An algorithm with these desirable properties is the MH algorithm described in detail in Sect. 5.6 (p. 311). The last row of Fig. B.2 displays the sampling paths for β_0 and β_1 and the autocorrelation function for β_1 . Obviously, this algorithm has the best properties among the four algorithms suggested so far. Autocorrelations are practically zero for lags about 13 and larger. Variances in the proposal density are chosen automatically and correlation of parameters is taken into account.

Using this algorithm a Markov chain was generated and, after the burn-in phase, 20,000 random numbers were drawn. Saving every 20th random number led to a thinned sample of size 1,000. Based on this thinned sample all characteristics of interest of the posterior can be approximated. To approximate the posterior mean we compute the arithmetic means for the sample, resulting in $\hat{\beta} = (-0.64, 0.65, -0.38)'$. This estimate is very close to the MLEs. Estimation of credible intervals can be based on the quantiles of the sampled random numbers. For example, we obtain 95 % credible intervals by choosing the 2.5 % quantiles as lower and 97.5 % quantiles as upper bounds. This results in the intervals $[-0.87, -0.42]$, $[0.52, 0.78]$, and $[-0.52, -0.26]$ for the sample generated in our example. They are also quite close to the confidence intervals of the regression coefficients obtained by likelihood inference (but have to be interpreted differently).

A further advantage of simulation-based inference is that inference is also easily possible for (complicated) nonlinear transformations of the original parameters. If, for example, we are interested in a credible interval for $\exp(\beta_0 + \beta_1 + \beta_2)$, then we simply compute the transformed quantity for the random numbers from the original sample. A 95 % credible interval for the transformed parameter is simply given by corresponding 2.5 % and 97.5 % quantiles.

△

2.2 Gibbs Sampler

Let $p(\boldsymbol{\theta} | \mathbf{y})$ be the posterior and assume that $\boldsymbol{\theta}$ is partitioned into S blocks $\boldsymbol{\theta}_1, \dots, \boldsymbol{\theta}_S$. The Gibbs sampler generates random numbers as follows:

1. Specify initial values $\boldsymbol{\theta}_1^{(0)}, \dots, \boldsymbol{\theta}_S^{(0)}$ and the number T of iterations. Set $t = 1$.
2. For $s = 1, \dots, S$, draw random numbers from the full conditionals

$$p(\boldsymbol{\theta}_s | \boldsymbol{\theta}_1^{(t)}, \dots, \boldsymbol{\theta}_{s-1}^{(t)}, \boldsymbol{\theta}_{s+1}^{(t-1)}, \dots, \boldsymbol{\theta}_S^{(t-1)}, \mathbf{y}).$$

Note that the most actual states are used in the conditioning set of parameter blocks.

3. Stop if $t = T$; otherwise set $t = t + 1$ and go to 2.

Gibbs Sampler and Hybrid Algorithms

In many practical applications the parameter vector is high-dimensional. The acceptance rates then become rather small even for well-designed MH algorithms, because a high-dimensional random number has to be accepted or not. So-called hybrid algorithms provide a solution to this problem, using a “divide and conquer” strategy. The high-dimensional parameter vector $\boldsymbol{\theta}$ is partitioned into smaller blocks $\boldsymbol{\theta}_1, \boldsymbol{\theta}_2, \dots, \boldsymbol{\theta}_S$. Separate MH steps are then constructed for these subvectors.

The *Gibbs sampler* is the simplest special case of this strategy. In most cases no (simple) methods for directly drawing random numbers from the density $p(\boldsymbol{\theta} | \mathbf{y})$ of the entire parameter vector are available. Often, however, random numbers can be directly drawn from the conditional densities $p(\boldsymbol{\theta}_1 | \cdot), p(\boldsymbol{\theta}_2 | \cdot), \dots, p(\boldsymbol{\theta}_S | \cdot)$, where $p(\boldsymbol{\theta}_s | \cdot)$ denotes the conditional density of $\boldsymbol{\theta}_s$ given all other blocks $\boldsymbol{\theta}_1, \dots, \boldsymbol{\theta}_{s-1}, \boldsymbol{\theta}_{s+1}, \dots, \boldsymbol{\theta}_S$ and the data $\mathbf{y}$. These densities are called full conditionals. Gibbs sampling consists in successively drawing random numbers from the full conditionals in each iteration and accepting them (with probability one) as the next state of the Markov chain; see Box 2.2 for a summary. After a burn-in phase the sampled random numbers can be considered as realizations from the marginal posteriors $p(\boldsymbol{\theta}_1 | \mathbf{y}), p(\boldsymbol{\theta}_2 | \mathbf{y}), \dots, p(\boldsymbol{\theta}_S | \mathbf{y})$.

If it is not possible to directly draw random numbers from some of the full conditionals, then an MH step is included instead. For the corresponding block $\boldsymbol{\theta}_s$, a proposal density

$$q_s(\boldsymbol{\theta}_s^* | \boldsymbol{\theta}_1^{(t)}, \dots, \boldsymbol{\theta}_{s-1}^{(t)}, \boldsymbol{\theta}_s^{(t-1)}, \dots, \boldsymbol{\theta}_S^{(t-1)})$$

is chosen and random numbers $\boldsymbol{\theta}_s^*$ are drawn from it. They are accepted as new states of the Markov chain with probability

$$\alpha(\theta_s^* | \theta_s^{(t-1)}) = \min \left\{ \frac{p(\theta_s^* | \theta_{-s}^{(t-1)}) q_s(\theta_s^{(t-1)} | \theta_1^{(t)}, \dots, \theta_{s-1}^{(t)}, \theta_s^*, \dots, \theta_S^{(t-1)})}{p(\theta_s^{(t-1)} | \theta_{-s}^{(t-1)}) q_s(\theta_s^* | \theta_1^{(t)}, \dots, \theta_{s-1}^{(t)}, \theta_s^{(t-1)}, \dots, \theta_S^{(t-1)})}, 1 \right\},$$

where $p(\theta_s | \theta_{-s}^{(t-1)}) = p(\theta_s | \theta_1^{(t)}, \dots, \theta_{s-1}^{(t)}, \theta_{s+1}^{(t-1)}, \dots, \theta_S^{(t-1)}, y)$ denotes the full conditional of θ_s . Otherwise, $\theta_s^{(t)} = \theta_s^{(t-1)}$ as in the original MH algorithm. In summary, step 2 of the Gibbs sampler is replaced by:

2.* For $s = 1, \dots, S$, draw random numbers θ_s^* from the proposal densities $q_s(\theta_s^* | \cdot)$ and accept them with probability $\alpha(\theta_s^* | \theta_s^{(t-1)})$; otherwise set $\theta_s^{(t)} = \theta_s^{(t-1)}$.

Note that MH and Gibbs steps in 2.* can be used in combination. Formally, the corresponding full conditional is directly chosen as proposal density. The acceptance probability then becomes $\alpha(\theta_s^* | \theta_s^{(t-1)}) = 1$, so that all proposed values are accepted.

B.5.4 Model Selection

The classical approach for Bayesian model choice is to compare competing models through the *posterior probabilities* of the models. Suppose we are given K competing models $M_1, \dots, M_K$ with associated parameters $\theta_1, \dots, \theta_K$. By a “model” we mean a set of probability distributions. For instance M_j , $j = 1, \dots, K$, could denote the regression models $y | \theta_j, M_j \sim N(X_j \theta_j, \sigma^2 I)$ (with known variance σ^2 for simplicity). For each model M_j let $p(y | \theta_j, M_j)$ denote the observation model and $p(\theta_j | M_j)$ be the prior for the model parameters θ_j . The posterior for θ_j under model M_j is then given by

$$p(\theta_j | y, M_j) = \frac{p(y | \theta_j, M_j) p(\theta_j | M_j)}{p(y | M_j)},$$

where

$$p(y | M_j) = \int p(y | \theta_j, M_j) p(\theta_j | M_j) d\theta_j \quad (\text{B.5})$$

is the *marginal likelihood*. For model selection, we additionally have to assign prior probabilities $p(M_j)$ associated with each model M_j . Now the competing models can be compared through the posterior model probabilities given by

$$p(M_j | y) = \frac{p(y | M_j) p(M_j)}{p(y)} \propto p(y | M_j) p(M_j)$$

with

$$p(y) = \sum_{k=1}^K p(y | M_k) p(M_k).$$

We prefer model M_j against model M_s if $p(M_j | \mathbf{y}) > p(M_s | \mathbf{y})$ or the posterior ratio

$$\frac{p(M_j | \mathbf{y})}{p(M_s | \mathbf{y})} = \frac{p(M_j)}{p(M_s)} \frac{p(\mathbf{y} | M_j)}{p(\mathbf{y} | M_s)}$$

is larger than 1. In cases of equal priors $p(M_1) = p(M_2) = \dots = p(M_K) = 1/K$ the posterior ratio simplifies to the *Bayes factor*

$$BF_{js}(\mathbf{y}) = \frac{p(\mathbf{y} | M_j)}{p(\mathbf{y} | M_s)}.$$

If none of the competing models is favored prior to analysis of the data, model choice is based on Bayes factors.

Care has to be taken when using the Bayes factor in combination with improper priors. Suppose, we assume improper priors

$$p(\boldsymbol{\theta}_j | M_j) \propto c_j, \quad p(\boldsymbol{\theta}_s | M_s) \propto c_s$$

for the model parameters $\boldsymbol{\theta}_j$ and $\boldsymbol{\theta}_s$ of models M_j and M_s . Here c_j and c_s are arbitrary constants. The Bayes factor then becomes

$$BF_{js}(\mathbf{y}) = \frac{c_j}{c_s} \frac{\int p(\mathbf{y} | \boldsymbol{\theta}_j, M_j) d\boldsymbol{\theta}_j}{\int p(\mathbf{y} | \boldsymbol{\theta}_s, M_s) d\boldsymbol{\theta}_s},$$

where c_j/c_s is an arbitrary constant. This in turn implies that the Bayes factor is not uniquely defined. Thus, improper priors cannot be used in the context of Bayesian model choice, at least if Bayes factors or in other words marginal likelihoods are involved. Improper priors may be appropriate only if the parameter $\boldsymbol{\theta}$ is the “same” under all models under consideration.

In many applications the exact computation of Bayes factors is difficult because computation of the marginal model likelihoods $p(\mathbf{y} | M_j)$ causes problems. An approximation (after multiplying with -2) is

$$-2p(\mathbf{y} | M_j) \approx -2 \cdot \log(p(\mathbf{y} | \hat{\boldsymbol{\theta}}_j, M_j)) + \log(n) p_j,$$

where p_j is the dimension of the parameter vector $\boldsymbol{\theta}_j$ and $\hat{\boldsymbol{\theta}}_j$ is the posterior mode. The approximation can be derived through a Laplace approximation of the integral in Eq. (B.5) and leads to the Bayesian information criterion (BIC). For a model with parameter $\boldsymbol{\theta}$, log-likelihood $l(\boldsymbol{\theta})$, and MLE $\hat{\boldsymbol{\theta}}$ the BIC is defined as

$$BIC = -2l(\hat{\boldsymbol{\theta}}) + \log(n) p.$$

Among a set of competing models, the model with the smallest BIC will be selected. Formally, the BIC is similar to the AIC: the factor 2 multiplying the number of parameters in the AIC is replaced by $\log(n)$. Derivation of the two criteria is quite

different, however. In general, the BIC selects less complex models than the AIC because penalization of the number of parameters is stronger.

Although derived from a Bayesian perspective, the BIC is not very popular in Bayesian inference. The main reasons are: First, the assumptions underlying the derivation of the BIC as an approximation of marginal log-likelihoods are not sufficiently well fulfilled in complex high-dimensional models. Related to this is the problem of determining n in the factor $\log n$. It is not always the data sample size: For example, in mixed models n is the number of individuals. Second, when more complex Bayesian models are fitted with MCMC methods, the BIC is not directly available anyway.

A more recent criterion for model choice that has become quite popular in connection with MCMC inference is the deviance information criterion (DIC). Refer to Spiegelhalter, Best, Carlin, and van der Linde (2002). Its popularity is due to the fact that it can be easily computed from MCMC output. Let $\theta^{(1)}, \dots, \theta^{(T)}$ denote an MCMC sample from the posterior of the model. Computation of the DIC is based on two quantities: The first is the (unstandardized) deviance

$$D(\theta) = -2 \log(p(y | \theta))$$

of the model. The second is the *effective number* p_D of parameters in the model. It can be estimated through

$$p_D = \overline{D(\theta)} - D(\bar{\theta}),$$

where

$$\overline{D(\theta)} = \frac{1}{T} \sum_{t=1}^T D(\theta^{(t)})$$

is the average posterior deviance and $D(\bar{\theta})$ is the deviance evaluated at $\bar{\theta} = \frac{1}{T} \sum_{t=1}^T \theta^{(t)}$. The DIC is then defined as

$$\text{DIC} = \overline{D(\theta)} + p_D = 2\overline{D(\theta)} - D(\bar{\theta}).$$

As a disadvantage, the DIC value changes for different MCMC random samples. Therefore it may happen that model choice by DIC can lead to selecting different models with different MCMC samples. This will be only the case, however, if the DIC values of the models are quite close. It then will be better to keep both (or several) competing models under consideration.

Example B.11 Bayesian Logit Model—DIC

We illustrate the use of DIC with the simulated data from Example B.10. If we mistakenly omit the covariate x_2 and fit a logit model with x_1 only, then we obtain the (estimated) values $p_D = 1.99$ and $\text{DIC} = 571.6$. The effective number of parameters of about 2 is plausible, because we have estimated exactly two parameters β_0 and β_1 . Fitting the correctly specified model, we obtain $p_D = 2.93$ and $\text{DIC} = 540.3$. The effective number of parameters is now about 3, as had to be expected. The DIC is now considerably smaller than for the wrong model so that the more complex, true model is selected.

For illustration, we fit the correct model with five further MCMC runs. For p_D we obtain the values 3.05, 2.99, 3.15, 2.87, and 3.23 and for the DIC 540.56, 540.42, 540.73, 540.19, and 540.91, respectively. We see that the DIC varies between the different MCMC runs, but variability is usually quite low.

△

B.5.5 Model Averaging

If the focus of the analysis is not necessarily on selecting a single best model, model averaging could be a promising alternative. Suppose we are interested in inference regarding a quantity Δ . If we perform model selection, we first determine the model M_* with highest posterior probability. The inference is then based on the posterior $p(\Delta | y, M_*)$ of Δ conditional on the “best” model M_* . Alternatively, inference could be based on the unconditional posterior $p(\Delta | y)$ which is given by

$$p(\Delta | y) = \sum_{k=1}^K p(\Delta | y, M_k) p(M_k | y).$$

Hence, $p(\Delta | y)$ is obtained as a simple average of the posteriors $p(\Delta | y, M_k)$ under model M_k weighted by the posterior model probabilities $p(M_k | y)$. Model averaging might be particularly favorable (in terms of predictive power) compared to selecting a single best model if the posterior probability mass is not concentrated on a particular model.